

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\workflows\wf_8d68d336-288\agent-ab0517c4fddfdb32a.jsonl`  
- SHA-256: `ceef7ff6e6f243c7117cab8c4fee1c337c5638bf5e81b33640f648610668d469`  
- Source modified: `2026-07-03T19:49:41+00:00`  
- Imported at: `2026-07-08T16:00:15+00:00`  
- project: `wf_8d68d336-288`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T19:48:21.201Z)

Repo (my in-progress optimization branch): C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/fix-worktree

Key files to Read:

- backend/pipeline/detectors/wasb_infer.py ? the inference loop (DataLoader ? detector.run_tensor ? tracker). Contains the streaming SequentialFrameStore, the _PrefetchIter I just added, build_windows, and the tracker re-run loop.
- backend/pipeline/detectors/native_wasb_runner.py ? builds the wasb_infer.py argv on the GPU box (native runner). Has device/stream_video/batch_size/prefetch_batches config.
- backend/config/models.py (WasbIndexConfig) and backend/config/presets.py (cloud-serving preset).
- deploy/cloudrun/Dockerfile ? the WASB conda env is torch 1.11.0+cu113 (py3.8), a SEPARATE env from the app's torch. The detector shells out to it as a subprocess.

The WASB model itself lives at /opt/models/WASB-SBDT (nttcom/WASB-SBDT) ? you can't read it, but reason from how wasb_infer.py calls build_detector(cfg) / detector.run_tensor(imgs, trans) and build_tracker(cfg).

MEASURED BASELINE (first real L4 cloud run, 2026-07-03): a ~7.5-min 60fps badminton clip took Phase-2 (WASB detection) 727.9s (~0.6x realtime); GPU utilization sat <1% the whole time. Hardware: NVIDIA L4 (Ada sm_89, 23GB), 8 vCPU, 32GB. Streaming mode forces DataLoader num_workers=0 (single-process), so CPU batch assembly (cv2 decode + PIL + ToTensor) serializes with GPU inference.

CRITICAL CONSTRAINT ? the D5/D16 byte-parity guarantee: the repo REQUIRES that the rally/window OUTPUT (the candidate windows and trajectory CSV) stay bit-identical / deterministic across runs and across the disk-frames vs streaming paths. This is asserted by a real parity test. An optimization that changes detector NUMERICS (FP16, TF32, cudnn.benchmark algo selection, NVDEC vs CPU decode) risks changing the output and breaking that guarantee. Batch size and prefetch do NOT change per-frame numerics (same forward, just grouped/overlapped) so they are parity-safe. Already implemented on this branch: indexing.wasb.batch_size + prefetch_batches knobs, the cloud-serving preset opts into batch 64 + prefetch 4.

You are a CUDA/PyTorch performance engineer. Investigate ONE optimization lever in depth by READING the actual code. Lever "decode-nvdec":
Investigate GPU-accelerated video decode (NVDEC) for the streaming path ? currently cv2.VideoCapture decodes on CPU (make_video_frame_loader). D16 explicitly DEFERRED NVDEC behind a config knob citing byte-parity (NVDEC vs CPU decode can differ per-pixel). Assess: would moving decode off the CPU free the CPU to feed batches faster (the real bottleneck is CPU-feed-bound)? Options: PyNvVideoCodec / torchaudio/torchvision GPU decode / ffmpeg NVDEC piping. Parity verdict and whether decode-side changes even reach the detector output given downstream resize/normalize.

Read the named files, reason from the real code, and return the structured finding. Be concrete about code_locations (file:function). Be honest about parity_risk ? the byte-parity guarantee is load-bearing.

2. assistant (2026-07-03T19:48:23.716Z)

I'll investigate the NVDEC decode lever by reading the actual code. Let me start with the key files.

3. user (2026-07-03T19:48:26.539Z)

```
1      """WASB shuttle-trajectory inference on an arbitrary frame sequence (runs in WSL).
2
3      WASB-SBDT only ships *dataset* evaluation; this is a thin wrapper that reuses its
4      own building blocks (ImageDataset transform + detector + online tracker) to run on
5      any folder of extracted frames and emit a trajectory CSV in the exact shape the
6      indexer's `tracknet_runner.parse_trajectory_csv` already consumes:
7
8          Frame,Visibility,X,Y
9
10     It must run INSIDE the WSL `wasb` conda env, from the WASB-SBDT `src/` dir (it
11     imports WASB's `detectors`/`trackers`/`dataloaders`). The Windows-side adapter
12     (`wasb_runner.py`) stages this file + the frames into WSL and invokes it.
13
14     Resumable across reboots
15     -----
16     The GPU detector pass (one forward per sliding window ? ~21k for a 6-min clip) is
17     the slow, expensive part; the CPU tracker pass that turns detections into a
18     trajectory is cheap and *stateful* (it must see every frame in order, so it can't
19     resume mid-stream). We exploit that split:
20
21     * The detector pass writes its per-window raw outputs incrementally to
22       ``<cache>/det_raw.jsonl`` (flushed+fsync'd every ``--flush-every`` windows) and
23       records progress in ``<cache>/manifest.json`` (atomic write). A reboot loses at
24       most ``--flush-every`` windows of GPU work instead of the whole pass.
25     * On restart we replay the cached windows, skip the DataLoader ahead to the first
26       un-cached window, and continue. Once every window is cached, the detector pass
27       is skipped entirely and we just re-run the (cheap, deterministic) tracker over
28       the full ordered cache -> trajectory CSV. So a lost trajectory CSV costs seconds,
29       not the GPU hour.
30
31     The cache is keyed by the output path (``<out>_wasbcache/`` by default), lives next
32     to the output (NOT in %TEMP%), and is invalidated automatically if the frames dir,
33     weights, sport, window size, or frame count change.
34
35     Usage (in WSL, from ~/models/WASB-SBDT/src):
36         python wasb_infer.py --frames_dir /path/frames --weights
37         ../pretrained_weights/wasb_badminton_best.pth.tar \
38         --out /path/traj.csv [--sport badminton] [--limit N] [--cache-dir DIR] [--flush-
39         every 1000] [--fresh]
40         """
41
42     import argparse
43     import csv
44     import glob
45     import json
46     import logging
47     import os
48     import os.path as osp
49     from collections import defaultdict
50     from contextlib import contextmanager
51
52     logger = logging.getLogger(__name__)
```

```
CACHE_VERSION = 1
```

```

53     _DET_FILE = "det_raw.jsonl"
54     _MANIFEST_FILE = "manifest.json"
55
56
57     def _build_cfg(weights: str, sport: str, device: str = "cuda"):
58         """Compose the WASB eval config via Hydra, overriding model/weights/device.
59
60         ``device`` is ``cuda`` (default ? the historical WSL behaviour, byte-parity
61         preserved),
62         ``mps`` (Apple) or ``cpu``. Only the single-GPU CUDA path requests
63         ``runner.gpus=[0]``;
64         cpu/mps run with no GPU list so the detector places tensors on ``device``.
65         """
66         from hydra import compose, initialize
67
68         gpus = "[0]" if device == "cuda" else "[]" # single GPU on cuda; none for cpu/mps
69         with initialize(config_path="configs", version_base=None):
70             cfg = compose(
71                 config_name="eval",
72                 overrides=[
73                     f"dataset={sport}",
74                     "model=wasb",
75                     f"detector.model_path={weights}",
76                     f"runner.device={device}",
77                     f"runner.gpus={gpus}", # default config assumes multiple GPUs; pin to
78                     this box
79                 ],
80             )
81             return cfg
82
83     def _frame_id(path: str) -> int:
84         """Best-effort integer frame id from a frame filename (e.g. frame_000180.png ->
85         180)."""
86         stem = osp.splitext(osp.basename(path))[0]
87         digits = "".join(ch for ch in stem if ch.isdigit())
88         return int(digits) if digits else -1
89
90     # ----- #
91     # Resumable detector cache (pure-Python, no torch/numpy ? unit-testable)
92     # ----- #
93     def _cache_paths(cache_dir: str):
94         return osp.join(cache_dir, _DET_FILE), osp.join(cache_dir, _MANIFEST_FILE)
95
96     def default_cache_dir(out_csv: str) -> str:
97         """Stable cache dir next to the trajectory output (survives reboots; not %TEMP%)."""
98         d = osp.dirname(osp.abspath(out_csv))
99         stem = osp.splitext(osp.basename(out_csv))[0]
100         return osp.join(d, stem + "_wasbcache")
101
102     def _atomic_write_text(path: str, text: str) -> None:
103         """Write then rename so a crash mid-write can't leave a half-written file."""
104         tmp = path + ".tmp"
105         with open(tmp, "w") as f:
106             f.write(text)
107             f.flush()
108             os.fsync(f.fileno())
109         os.replace(tmp, path)
110
111     def _det_plain(det) -> list:
112         """A detector candidate dict {'xy': np.array([x,y]), 'score', 'scale'} -> JSON-able

```

```

list."""
114     xy = det["xy"]
115     return [float(xy[0]), float(xy[1]), float(det["score"]), int(det["scale"])]
116
117
118     def _dets_to_tracker(plain_list):
119         """Inverse of _det_plain: rebuild the dicts the tracker expects (xy MUST be a numpy
array,
120         the tracker does vector math / np.linalg.norm on it)."""
121         import numpy as np
122
123         return [
124             {"xy": np.array([p[0], p[1]], dtype=float), "score": p[2], "scale": p[3]}
125             for p in plain_list
126         ]
127
128
129     def write_manifest(cache_dir: str, manifest: dict) -> None:
130         _, mpath = _cache_paths(cache_dir)
131         _atomic_write_text(mpath, json.dumps(manifest, indent=2))
132
133
134     def read_manifest(cache_dir: str):
135         _, mpath = _cache_paths(cache_dir)
136         if not osp.exists(mpath):
137             return None
138         try:
139             with open(mpath) as f:
140                 return json.load(f)
141         except (json.JSONDecodeError, OSError):
142             return None
143
144
145     def manifest_compatible(
146         manifest: dict,
147         *,
148         frames_dir: str,
149         weights: str,
150         sport: str,
151         frames_in: int,
152         frames_total: int,
153         device: str = "cuda",
154     ) -> bool:
155         """A cache is reusable only if the run that produced it matches this run's inputs.
156
157         ``device`` defaults to ``cuda`` so caches written before device-keying (no
158         ``device``
159         field) stay compatible with a CUDA run ? but a cpu/mps run will NOT reuse a cuda
160         cache
161         (detections can differ numerically across devices), and vice-versa.
162
163         ``frames_dir`` is the **already-canonical cache key** the caller stored in the
164         manifest
165         (``run()`` passes its ``cache_key`` here, and writes the same value into the
166         manifest):
167         ``osp.abspath(frames_dir)`` for the frames-on-disk path, or the synthetic
168         ``video:<abspath>``
169         key for the streaming path. We therefore compare it VERBATIM ? do NOT re-``abspath``
170         it here:
171         ``osp.abspath("video:/x.mp4")`` is not recognised as absolute and gets the CWD
172         prepended,
173         so a streaming cache would never match itself and every re-run would recompute
174         (issue #348).
175         """
176         if not manifest or manifest.get("version") != CACHE_VERSION:

```

```

169         return False
170     return (
171         manifest.get("frames_dir") == frames_dir
172         and manifest.get("weights") == weights
173         and manifest.get("sport") == sport
174         and manifest.get("frames_in") == frames_in
175         and manifest.get("frames_total") == frames_total
176         and manifest.get("device", "cuda") == device
177     )
178
179
180 def reset_cache(cache_dir: str) -> None:
181     """Drop any existing cache so the next run starts clean."""
182     det_jsonl, mpath = _cache_paths(cache_dir)
183     for p in (det_jsonl, mpath):
184         if osp.exists(p):
185             os.remove(p)
186
187
188 def load_and_clean_cache(cache_dir: str):
189     """Read the longest *contiguous* (w == line index) valid prefix of det_raw.jsonl,
190     rebuild the per-frame detection accumulation, and rewrite the file to exactly that
191     prefix so a trailing half-written line from a crash can't corrupt future appends.
192
193     Returns (windows_done, det_by_fid) where det_by_fid maps frame_id -> list of
194     [x, y, score, scale] candidates (accumulated across every window the frame is in,
195     in window order ? identical to a non-resumed run).
196     """
197     det_jsonl, _ = _cache_paths(cache_dir)
198     det_by_fid = defaultdict(list)
199     valid: list = []
200     if osp.exists(det_jsonl):
201         with open(det_jsonl, "r") as f:
202             for line in f:
203                 s = line.strip()
204                 if not s:
205                     continue
206                 try:
207                     obj = json.loads(s)
208                 except json.JSONDecodeError:
209                     break # truncated trailing line (crash mid-flush)
210                 if obj.get("w") != len(valid): # non-contiguous -> stop at the gap
211                     break
212                 valid.append(s)
213                 for fid, plain in obj["f"]:
214                     det_by_fid[int(fid)].extend([list(p) for p in plain])
215                 # Guarantee a clean append boundary by rewriting only the good prefix.
216                 _atomic_write_text(det_jsonl, ("\n".join(valid) + "\n") if valid else "")
217     return len(valid), det_by_fid
218
219
220 def _append_windows(fh, objs) -> None:
221     """Append window records as JSONL and durably flush (bounded loss on crash)."""
222     for o in objs:
223         fh.write(json.dumps(o, separators=(",", ":")) + "\n")
224     fh.flush()
225     os.fsync(fh.fileno())
226
227
228 def _atomic_write_trajectory(out_csv: str, rows) -> None:
229     os.makedirs(osp.dirname(osp.abspath(out_csv)), exist_ok=True)
230     tmp = out_csv + ".tmp"
231     with open(tmp, "w", newline="") as f:
232         w = csv.writer(f)
233         w.writerow(["Frame", "Visibility", "X", "Y"])

```

```

234         w.writerows(rows)
235         f.flush()
236         os.fsync(f.fileno())
237     os.replace(tmp, out_csv)
238
239
240     # ----- #
241     # Frame addressing + windowing (pure; no torch/cv2 ? unit-testable)
242     # ----- #
243     _STREAM_FRAME_FMT = "{:08d}.png"
244
245
246     def synthetic_frame_path(index: int) -> str:
247         """Stable, index-encoding frame name used by the streaming path.
248
249         It mirrors the on-disk names ``extract_frames`` writes (``{i:08d}.png``) so the
250         downstream integer-id parser (``_frame_id``) yields the SAME frame id whether the
251         frames came from disk or from the in-memory stream. The streaming image loader
252         inverts this name back to an index to fetch the decoded frame.
253         """
254         return _STREAM_FRAME_FMT.format(int(index))
255
256
257     def build_windows(frames: list, frames_in: int):
258         """Sliding windows of ``frames_in`` consecutive frame *paths* (the unit of GPU work
259         + checkpoint). Pure list math, identical for the disk and streaming paths.
260
261         Returns a list of windows, each a list of ``frames_in`` consecutive paths
262         (``[frames[i:i+frames_in] for i in range(len(frames) - frames_in + 1)]``). The
263         caller wraps each window into the ``{"frames", "annos"}`` sample shape WASB's
264         ``ImageDataset`` expects; keeping that out of here stays torch-free for unit tests.
265         """
266         return [frames[i : i + frames_in] for i in range(len(frames) - frames_in + 1)]
267
268
269     class SequentialFrameStore:
270         """In-memory sliding buffer over a forward-only frame reader, sized for the
271         detector's sliding-window access pattern (peak O(window) frames, not O(video)).
272
273         The detector DataLoader runs with ``shuffle=False`` and (in streaming mode)
274         ``num_workers=0``. ``ImageDataset.__getitem__(i)`` reads the frames of window ``i``
275         in order ? ``i, i+1, ..., i+window-1`` ? and samples are requested ``i = start,
276         start+1, ...``. So the global read sequence dips back by ``window-1`` at each window
277         boundary (``0,1,2, 1,2,3, 2,3,4, ...`` for ``window=3``); the highest index seen so
278         far never decreases. We keep only frames at or above ``max_seen - (window-1)`` (the
279         earliest index any not-yet-finished window can still ask for) and decode each source
280         frame exactly once via the forward-only ``reader(index)``.
281         """
282
283         def __init__(self, reader, total: int, window: int = 1, start_index: int = 0):
284             self._reader = reader
285             self._total = int(total)
286             self._window = max(1, int(window))
287             self._buf: dict = {}
288             # On a resumed run the detector's first needed frame is `start_index`; begin
289             # pulling there so the reader can skip (seek past) the already-cached prefix
290             # instead of decoding 0..start_index-1 just to throw them away.
291             self._next = int(start_index) # next index not yet pulled from the reader
292             self._max_seen = int(start_index) - 1
293             self._floor = int(start_index) # lowest index we still keep (never evict below)
294
295         def get(self, index: int):
296             index = int(index)
297             if index < self._floor:
298                 raise KeyError(

```

```

299         f"frame {index} already evicted (below floor {self._floor}); the access
300         "
301         f"pattern must stay within a window of the highest frame seen"
302     )
303     if index >= self._total:
304         raise KeyError(f"frame {index} out of range (total {self._total})")
305     # Pull forward from the reader until the requested index is buffered.
306     while self._next <= index:
307         self._buf[self._next] = self._reader(self._next)
308         self._next += 1
309     if index > self._max_seen:
310         self._max_seen = index
311     # Evict frames older than the earliest a still-running window could re-request:
312     # once we've seen index `m`, no future window starts before `m - (window-1)`.
313     new_floor = max(self._floor, self._max_seen - (self._window - 1))
314     for k in range(self._floor, new_floor):
315         self._buf.pop(k, None)
316     self._floor = new_floor
317     return self._buf[index]
318
319 def _index_from_synthetic_path(path: str) -> int:
320     """Invert ``synthetic_frame_path``: a frame path -> its integer source index.
321
322     Reuses ``_frame_id`` (digits-only parse) so the index and the cached frame-id stay
323     in lockstep with the on-disk naming scheme.
324     """
325     return _frame_id(path)
326
327
328 def _bgr_to_pil_rgb(frame_bgr):
329     """Convert an in-memory BGR uint8 frame (as ``cv2.VideoCapture`` yields) to the
330     EXACT
331     PIL RGB image WASB's disk path produces.
332
333     The disk path is ``frame_bgr -> cv2.imwrite(PNG) ->
334     Image.open(PNG).convert('RGB')``;
335     because PNG is lossless that round-trip equals ``cv2.cvtColor(frame_bgr,
336     COLOR_BGR2RGB)``
337     bit-for-bit (verified empirically, max|diff|=0). Returning that here makes the
338     streamed
339     frame indistinguishable from the on-disk one to everything downstream.
340     """
341     import cv2
342     from PIL import Image
343
344     return Image.fromarray(cv2.cvtColor(frame_bgr, cv2.COLOR_BGR2RGB))
345
346
347 def _make_streamed_read_image(store, real_read_image):
348     """Build the ``read_image`` replacement used during a streaming detector pass.
349
350     A synthetic frame path -> the in-memory decoded frame for its index (as a PIL RGB
351     image,
352     matching ``read_image``'s real return type); any path that doesn't parse to an index
353     falls
354     through to ``real_read_image``. Pure ? imports no WASB module ? so it is unit-
355     testable
356     without the WSL-only ``dataloaders`` package.
357     """
358
359     def _read_image(path, *args, **kwargs):
360         idx = _index_from_synthetic_path(path)
361         if idx < 0:
362             return real_read_image(path, *args, **kwargs)

```

```

356         return _bgr_to_pil_rgb(store.get(idx))
357
358     return _read_image
359
360
361     @contextmanager
362     def _streaming_read_image_patch(
363         frame_loader, total: int, window: int = 1, start_index: int = 0
364     ):
365         """Scope a shim over WASB's ``read_image`` so ``ImageDataset`` is fed in-memory
366         decoded
367         frames during the detector pass, byte-for-byte as it would over real PNGs.
368
369         WASB's ``ImageDataset.__getitem__`` reads each frame via ``read_image(path)`` ? NOT
370         ``cv2.imread``. ``read_image`` (``utils.utils``) does
371         ``Image.open(path).convert('RGB')``,
372         returning a PIL **RGB** image, after an ``osp.exists(path)`` guard. We therefore
373         feed it a
374         PIL RGB image built from the streamed BGR frame (see ``_bgr_to_pil_rgb``), which is
375         byte-identical to the disk round-trip, and the shim never touches the filesystem so
376         the
377         ``osp.exists`` guard is sidestepped for synthetic stream paths.
378
379         We patch ``dataloaders.dataset_loader.read_image`` ? the binding the consumer
380         actually
381         calls (``dataset_loader`` does ``from utils import read_image``, so the name lives
382         in that
383         module's namespace; patching ``utils.read_image`` would NOT rebind the already-
384         imported
385         reference).
386
387         ``frame_loader(index) -> BGR uint8 ndarray`` is a forward-only sequential decoder
388         wrapped
389         in a ``SequentialFrameStore`` (sliding buffer sized to ``window`` = ``frames_in``);
390         on a
391         resume we start at ``start_index`` so the decoder seeks past already-cached windows.
392         The
393         original reader is restored on exit.
394
395         """
396         from dataloaders import dataset_loader as _dl
397
398         store = SequentialFrameStore(
399             frame_loader, total, window=window, start_index=start_index
400         )
401         real_read_image = _dl.read_image
402         # Rebind read_image in the consuming module for the duration of the detector pass;
403         # restore on exit. (Plain assignment, not setattr-with-constant, to stay ruff
404         B010-clean.)
405         _dl.read_image = _make_streamed_read_image(store, real_read_image) # type:
406         ignore[assignment]
407         try:
408             yield store
409         finally:
410             _dl.read_image = real_read_image # type: ignore[assignment]
411
412
413     # ----- #
414     # Inference
415     # ----- #
416     class _PrefetchIter:
417         """Bounded producer-thread wrapper over a batch iterator (the num_workers=0
418         DataLoader).
419
420         In streaming mode the DataLoader must stay single-process (the in-memory frame store
421         +

```



```

407     the cv2.imread shim live in this process and cannot be pickled to workers), so the
408     CPU-side batch assembly (decode + PIL + ToTensor) SERIALIZES with GPU inference ?
409     measured on the first real L4 serving run (2026-07-03) this left the GPU <1%
utilized.
410     A single producer THREAD fixes the overlap without multiprocessing: cv2/PIL/numpy
411     release the GIL for the heavy parts, the store's forward-only access pattern is
412     preserved (one consumer thread iterating sequentially), and the bounded queue caps
413     memory at ``depth`` batches. A producer exception is re-raised in the consumer."""
414
415     _END = object()
416
417     def __init__(self, src, depth: int = 2):
418         import queue
419         import threading
420
421         self._q: "queue.Queue" = queue.Queue(maxsize=max(1, int(depth)))
422         self._exc: "BaseException | None" = None
423
424         def _fill() -> None:
425             try:
426                 for item in src:
427                     self._q.put(item)
428             except BaseException as e: # noqa: BLE001 - surfaced in __next__
429                 self._exc = e
430             finally:
431                 self._q.put(self._END)
432
433         self._t = threading.Thread(target=_fill, name="wasb-prefetch", daemon=True)
434         self._t.start()
435
436     def __iter__(self):
437         return self
438
439     def __next__(self):
440         item = self._q.get()
441         if item is self._END:
442             if self._exc is not None:
443                 raise self._exc
444             raise StopIteration
445         return item
446
447
448     def run(
449         frames,
450         weights: str,
451         out_csv: str,
452         sport: str = "badminton",
453         limit: int = 0,
454         batch_size: int = 8,
455         num_workers: int = 4,
456         log_every_batches: int = 50,
457         cache_dir: "str | None" = None,
458         flush_every: int = 1000,
459         fresh: bool = False,
460         frame_loader=None,
461         source_key: "str | None" = None,
462         device: str = "cuda",
463         prefetch_batches: int = 0,
464     ) -> str:
465         """Run WASB detection+tracking over an ordered frame source -> trajectory CSV.
466
467         Two equivalent ways to supply pixels (output is bit-identical between them):
468
469         * **frames-on-disk** (default): ``frames`` is a directory path string; frames are
470         globbed (``*.png``/``*.jpg``) and ``ImageDataset`` reads each file via

```

```

``cv2.imread``.
471     * **streaming** (fast path): ``frames`` is a pre-built ordered list of (synthetic)
472     frame paths and ``frame_loader(index) -> BGR uint8 ndarray`` supplies the decoded
473     pixels in-memory. We scope a shim over WASB's ``read_image`` (the actual per-frame
474     reader, which returns a PIL RGB image) over the DataLoader pass so every other
line of
475     ``ImageDataset.__getitem__`` runs UNCHANGED ? the tensor the detector sees is
identical
476     to the disk round-trip (``cv2.imread`` PNG -> ``Image.open().convert('RGB')`` ==
477     ``cvtColor(bgr, BGR2RGB)``), verified byte-identical). Streaming forces
``num_workers=0``
478     (the in-process shim + buffer can't cross worker processes).
479
480     ``source_key`` overrides the cache-keying identity (defaults to the abspath of the
481     frames dir); the streaming path passes a stable ``video:<abspath>`` key so a resume
482     after reboot still matches.
483     """
484     import time
485
486     import torch
487     from dataloaders import build_img_transforms, build_seq_transforms
488     from dataloaders.dataset_loader import ImageDataset
489     from torch.utils.data import DataLoader
490
491     # NB: this is WASB's OWN local `trackers` module (from its repo, on sys.path inside
the
492     # `wasb` conda env), NOT the Roboflow `trackers` package. The `type: ignore`
suppresses
493     # the mypy false positive where the main env resolves `trackers` to the Roboflow
package
494     # (which has no `build_tracker`). There is NO runtime collision ? this runs in a
separate
495     # subprocess/conda env. Do NOT "clean up" the ignore without removing the Roboflow
dep.
496     from trackers import build_tracker # type: ignore[attr-defined]
497     from utils import Center
498
499     from detectors import build_detector
500
501     streaming = frame_loader is not None
502
503     cfg = _build_cfg(weights, sport, device)
504     frames_in = int(cfg["model"]["frames_in"])
505     input_wh = (int(cfg["model"]["inp_width"]), int(cfg["model"]["inp_height"]))
506     output_wh = (int(cfg["model"]["out_width"]), int(cfg["model"]["out_height"]))
507
508     if streaming:
509         # `frames` is already the ordered list of (synthetic) frame paths.
510         if not isinstance(frames, (list, tuple)):
511             raise SystemExit(
512                 "streaming mode requires `frames` to be a list of frame paths"
513             )
514         frames = list(frames)
515         frames_dir = source_key or "stream"
516     else:
517         frames_dir = frames
518         frames = sorted(
519             glob.glob(osp.join(frames_dir, "*.png"))
520             + glob.glob(osp.join(frames_dir, "*.jpg"))
521         )
522     if limit:
523         frames = frames[:limit]
524     if len(frames) < frames_in:
525         where = frames_dir if not streaming else "video stream"
526         raise SystemExit(f"need >= {frames_in} frames, found {len(frames)} in {where}")

```

```

527
528     # Sliding windows of `frames_in` consecutive frames (the unit of GPU work +
checkpoint).
529     samples = []
530     for window in build_windows(frames, frames_in):
531         annos = [
532             {"frame_path": p, "center": Center(is_visible=False, x=-1.0, y=-1.0)}
533             for p in window
534         ]
535         samples.append({"frames": window, "annos": annos})
536     windows_total = len(samples)
537
538     # ---- cache setup / resume decision ----- #
539     if cache_dir is None:
540         cache_dir = default_cache_dir(out_csv)
541         os.makedirs(cache_dir, exist_ok=True)
542         det_jsonl, _ = _cache_paths(cache_dir)
543         cache_key = source_key if source_key is not None else osp.abspath(frames_dir)
544         manifest = None if fresh else read_manifest(cache_dir)
545         resume = manifest_compatible(
546             manifest or {},
547             frames_dir=cache_key,
548             weights=weights,
549             sport=sport,
550             frames_in=frames_in,
551             frames_total=len(frames),
552             device=device,
553         )
554         if resume:
555             windows_done, det_by_fid = load_and_clean_cache(cache_dir)
556             logger.info(
557                 f"resuming from cache: {windows_done}/{windows_total} windows already
detected"
558             )
559         else:
560             if manifest is not None:
561                 logger.info("cache incompatible with this run -> starting fresh")
562                 reset_cache(cache_dir)
563                 windows_done, det_by_fid = 0, defaultdict(list)
564
565         base_manifest = {
566             "version": CACHE_VERSION,
567             "frames_dir": cache_key,
568             "weights": weights,
569             "sport": sport,
570             "frames_in": frames_in,
571             "frames_total": len(frames),
572             "device": device,
573             "windows_total": windows_total,
574             "windows_done": windows_done,
575         }
576         write_manifest(cache_dir, base_manifest)
577
578     # ---- detector pass over the un-cached windows ----- #
579     remaining = samples[windows_done:]
580     if remaining:
581         _, transform_test = build_img_transforms(cfg)
582         try:
583             _, seq_transform_test = build_seq_transforms(cfg)
584         except Exception:
585             seq_transform_test = None
586         ds = ImageDataset(
587             cfg,
588             remaining,
589             input_wh,

```

```

590         output_wh,
591         transform=transform_test,
592         seq_transform=seq_transform_test,
593         is_train=False,
594     )
595     # Streaming forces single-process loading: the in-memory frame store + the
596     # cv2.imread shim live in THIS process and can't be pickled to DataLoader
workers.
597     loader_workers = 0 if streaming else num_workers
598     loader = DataLoader(
599         ds, batch_size=batch_size, shuffle=False, num_workers=loader_workers
600     )
601     detector = build_detector(cfg)
602
603     from contextlib import ExitStack
604
605     t0 = time.time()
606     done = windows_done
607     win_idx = windows_done
608     log_every = max(1, log_every_batches)
609     flush_n = max(1, flush_every)
610     pending = []
611     logger.info(
612         f"detecting on {len(remaining)} remaining windows "
613         f"(total {windows_total}, batch={batch_size}, workers={loader_workers}, "
614         f"source={'video-stream' if streaming else 'frames-dir'})..."
615     )
616     det_f = open(det_jsonl, "a")
617     try:
618         with ExitStack() as stack:
619             stack.enter_context(torch.no_grad())
620             # In streaming mode, route ImageDataset's per-frame `cv2.imread(path)`
to the
621             # in-memory decoded frame for that synthetic path; every other line of
622             # __getitem__ runs unchanged so the tensor matches the PNG round-trip
exactly.
623             # The detector's first needed frame is `windows_done` (window i starts
at
624             # frame i), so prime the store there for a resume.
625             if streaming:
626                 stack.enter_context(
627                     _streaming_read_image_patch(
628                         frame_loader,
629                         len(frames),
630                         window=frames_in,
631                         start_index=windows_done,
632                     )
633                 )
634             # Overlap CPU batch assembly with GPU inference (streaming/workers=0
only ?
635             # a multi-worker DataLoader already overlaps via its worker processes).
636             batches = (
637                 _PrefetchIter(loader, depth=prefetch_batches)
638                 if prefetch_batches and loader_workers == 0
639                 else loader
640             )
641             for bi, batch in enumerate(batches):
642                 imgs, _hms, trans = batch[0], batch[1], batch[2]
643                 img_paths = [
644                     list(t) for t in batch[-1]
645                 ] # [frame_in][batch] -> path
646                 batch_results, _ = detector.run_tensor(imgs, trans)
647                 nb = imgs.shape[0]
648                 for ib in range(nb):
649                     frames_payload = []

```

```

650         for ie in sorted(batch_results[ib].keys()):
651             p = img_paths[ie][ib]
652             fid = _frame_id(p)
653             plain = [_det_plain(d) for d in batch_results[ib][ie]]
654             frames_payload.append([fid, plain])
655             det_by_fid[fid].extend(plain)
656         pending.append({"w": win_idx, "f": frames_payload})
657         win_idx += 1
658     done += nb
659     if len(pending) >= flush_n or done >= windows_total:
660         _append_windows(det_f, pending)
661         pending = []
662         base_manifest["windows_done"] = done
663         write_manifest(cache_dir, base_manifest)
664     if (bi + 1) % log_every == 0 or done >= windows_total:
665         el = time.time() - t0
666         new_done = done - windows_done
667         rate = new_done / el if el > 0 else 0.0
668         eta = (windows_total - done) / rate if rate > 0 else 0.0
669         logger.info(
670             f"{done}/{windows_total} "
671             f"({100 * done // max(1, windows_total)}%) | {rate:.1f}
win/s | "
672             f"elapsed {el:.0f}s | ETA {eta:.0f}s"
673         )
674     if pending:
675         _append_windows(det_f, pending)
676         base_manifest["windows_done"] = done
677         write_manifest(cache_dir, base_manifest)
678     finally:
679         det_f.close()
680         logger.info(f"detection done in {time.time() - t0:.0f}s; running tracker...")
681 else:
682     logger.info("detector cache already complete -> tracker-only re-run")
683
684 # ---- tracker re-run over the full ordered cache (cheap, deterministic) ---- #
685 tracker = build_tracker(cfg)
686 tracker.refresh()
687 rows = []
688 for fid in sorted(det_by_fid.keys()):
689     r = tracker.update(_dets_to_tracker(det_by_fid[fid]))
690     rows.append((fid, 1 if r.get("visi") else 0, r.get("x", -1), r.get("y", -1)))
691
692 _atomic_write_trajectory(out_csv, rows)
693 visible = sum(1 for _, v, *_ in rows if v)
694 logger.info(f"wrote {len(rows)} rows -> {out_csv}")
695 logger.info(f"visible-shuttle frames: {visible}/{len(rows)}")
696 return out_csv
697
698
699 def extract_frames(video_path: str, frames_dir: str) -> str:
700     """Decode a video to PNG frames named by source frame index (00000000.png ...).
701
702     Idempotent: if the dir already holds the expected number of frames (matching the
703     container's reported frame count) we skip re-decoding ? re-extraction after a
704     reboot is wasteful. NOTE: cv2 PNG extraction is slow; for big clips prefer
705     extracting with ffmpeg on the host and passing --frames_dir.
706     """
707     import cv2
708
709     os.makedirs(frames_dir, exist_ok=True)
710     cap = cv2.VideoCapture(video_path)
711     if not cap.isOpened():
712         raise SystemExit(f"cannot open video: {video_path}")
713     expected = int(cap.get(cv2.CAP_PROP_FRAME_COUNT)) or 0

```

```

714     existing = len(glob.glob(osp.join(frames_dir, "*.png")))
715     if expected > 0 and existing >= expected:
716         cap.release()
717         logger.info(f"reusing {existing} already-extracted frames -> {frames_dir}")
718         return frames_dir
719     i = 0
720     while True:
721         ok, frame = cap.read()
722         if not ok:
723             break
724         cv2.imwrite(osp.join(frames_dir, f"{i:08d}.png"), frame)
725         i += 1
726     cap.release()
727     logger.info(f"extracted {i} frames -> {frames_dir}")
728     return frames_dir
729
730
731     def _probe_frame_count(video_path: str) -> int:
732         """Container-reported frame count via cv2 (0 if unknown). Used to size the
733         stream."""
734         import cv2
735         cap = cv2.VideoCapture(video_path)
736         if not cap.isOpened():
737             raise SystemExit(f"cannot open video: {video_path}")
738         n = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
739         cap.release()
740         return n
741
742
743     def make_video_frame_loader(video_path: str):
744         """Return ``(frame_loader, count_hint, release)`` for output-preserving streaming.
745
746         ``frame_loader(index) -> BGR uint8 ndarray`` decodes the video forward-only with one
747         long-lived ``cv2.VideoCapture``. It MUST be called with non-decreasing indices (the
748         detector's sliding-window order); it reads sequentially and only uses a frame-seek
749         to
750         skip forward when a *resume* starts past the current position. Each ``cap.read()``
751         returns exactly the BGR uint8 array that ``cv2.imwrite``/``cv2.imread`` of a PNG
752         would
753         yield (PNG is lossless), so the detector sees identical pixels to the disk path.
754
755         Returns the count hint from the container so the caller can build the frame list;
756         ``release`` closes the capture.
757         """
758         import cv2
759         cap = cv2.VideoCapture(video_path)
760         if not cap.isOpened():
761             raise SystemExit(f"cannot open video: {video_path}")
762         count_hint = int(cap.get(cv2.CAP_PROP_FRAME_COUNT) or 0)
763         state = {"pos": 0} # next index cap.read() will return
764
765         def frame_loader(index: int):
766             index = int(index)
767             if index < state["pos"]:
768                 raise RuntimeError(
769                     f"streaming decode is forward-only; got index {index} after
770                     {state['pos']}")
771             if index > state["pos"]:
772                 # Forward skip only happens when resuming past cached windows. Seek once.
773                 cap.set(cv2.CAP_PROP_POS_FRAMES, index)
774                 state["pos"] = index
775             ok, frame = cap.read()

```

```

775         if not ok:
776             raise RuntimeError(f"failed to decode frame {index} from {video_path}")
777         state["pos"] += 1
778         return frame
779
780     def release():
781         cap.release()
782
783     return frame_loader, count_hint, release
784
785
786 def run_video_streaming(
787     video_path: str,
788     weights: str,
789     out_csv: str,
790     sport: str = "badminton",
791     limit: int = 0,
792     batch_size: int = 8,
793     log_every_batches: int = 50,
794     cache_dir: "str | None" = None,
795     flush_every: int = 1000,
796     fresh: bool = False,
797     device: str = "cuda",
798     prefetch_batches: int = 0,
799 ) -> str:
800     """Fast path: decode ``video_path`` on the fly and feed frames straight to the
801     detector, WITHOUT extracting every frame to a PNG first (the disk round-trip that
802     dominates a long clip). Output is bit-identical to the frames-on-disk path; see
803     ``run``'s docstring for why.
804     """
805     frame_loader, count_hint, release = make_video_frame_loader(video_path)
806     try:
807         if count_hint <= 0:
808             raise SystemExit(
809                 f"could not determine frame count for {video_path}; cannot stream "
810                 f"(fall back to frame extraction with --frames_dir)"
811             )
812         frame_list = [synthetic_frame_path(i) for i in range(count_hint)]
813         source_key = "video:" + osp.abspath(video_path)
814         return run(
815             frame_list,
816             weights,
817             out_csv,
818             sport,
819             limit,
820             batch_size=batch_size,
821             num_workers=0,
822             log_every_batches=log_every_batches,
823             cache_dir=cache_dir,
824             flush_every=flush_every,
825             fresh=fresh,
826             frame_loader=frame_loader,
827             source_key=source_key,
828             device=device,
829             prefetch_batches=prefetch_batches,
830         )
831     finally:
832         release()
833
834
835 def main():
836     ap = argparse.ArgumentParser()
837     ap.add_argument("--frames_dir", help="folder of pre-extracted frames")
838     ap.add_argument(
839         "--video", help="video file; frames are extracted internally (in WSL)"
840     )

```

```

840     )
841     ap.add_argument(
842         "--frames_out_dir", help="where to extract frames when --video is used"
843     )
844     ap.add_argument(
845         "--stream-video",
846         action="store_true",
847         help="FAST PATH: decode --video on the fly and feed frames straight to "
848         "the detector (no PNG extraction). Output-equivalent to the disk path.",
849     )
850     ap.add_argument("--weights", required=True)
851     ap.add_argument("--out", required=True)
852     ap.add_argument("--sport", default="badminton")
853     ap.add_argument(
854         "--device",
855         default="cuda",
856         choices=["cuda", "mps", "cpu"],
857         help="torch device for the detector (default cuda = the WSL parity path)",
858     )
859     ap.add_argument(
860         "--limit", type=int, default=0, help="cap #frames (for quick tests)"
861     )
862     ap.add_argument("--batch-size", type=int, default=8)
863     ap.add_argument("--num-workers", type=int, default=4)
864     ap.add_argument(
865         "--prefetch-batches",
866         type=int,
867         default=0,
868         help="streaming mode: overlap CPU batch assembly with GPU inference via a "
869         "bounded producer thread of this queue depth (0 = off, the parity default)",
870     )
871     ap.add_argument(
872         "--log-every-batches", type=int, default=50, help="progress log cadence"
873     )
874     ap.add_argument(
875         "--cache-dir",
876         default=None,
877         help="resumable detector cache dir (default: <out>_wasbcache next to --out)",
878     )
879     ap.add_argument(
880         "--flush-every",
881         type=int,
882         default=1000,
883         help="fsync the detector cache every N windows (bounds reboot loss)",
884     )
885     ap.add_argument(
886         "--fresh", action="store_true", help="ignore any existing cache and start over"
887     )
888     args = ap.parse_args()
889
890     if args.stream_video:
891         if not args.video:
892             raise SystemExit("--stream-video requires --video")
893         run_video_streaming(
894             args.video,
895             args.weights,
896             args.out,
897             args.sport,
898             args.limit,
899             batch_size=args.batch_size,
900             log_every_batches=args.log_every_batches,
901             cache_dir=args.cache_dir,
902             flush_every=args.flush_every,
903             fresh=args.fresh,
904             device=args.device,

```



```

905         prefetch_batches=args.prefetch_batches,
906     )
907     return
908
909     frames_dir = args.frames_dir
910     if args.video:
911         frames_dir = extract_frames(
912             args.video, args.frames_out_dir or (args.video + "_frames")
913         )
914     if not frames_dir:
915         raise SystemExit("provide --frames_dir or --video")
916     run(
917         frames_dir,
918         args.weights,
919         args.out,
920         args.sport,
921         args.limit,
922         batch_size=args.batch_size,
923         num_workers=args.num_workers,
924         log_every_batches=args.log_every_batches,
925         cache_dir=args.cache_dir,
926         flush_every=args.flush_every,
927         fresh=args.fresh,
928         device=args.device,
929     )
930
931
932 if __name__ == "__main__":
933     main()
934

```

4. user (2026-07-03T19:48:27.147Z)

```

1     """Linux-native WASB shuttle-trajectory runner (no WSL).
2
3     The de-risk goal the ``DetectorRunner`` ABC was built for (see ``base.py`` ?
4     "Optional Linux-native path"; ``docs/DECOUPLED_COMPUTE/04-BUILD-AND-TEST-PLAN.md``
M3.5).
5     It is the sibling of :class:`~backend.pipeline.detectors.wasb_runner.WasbRunner` for a
6     Linux GPU box (GCP / a SaaS worker): it runs the same verified ``wasb_infer.py``
core
7     that the WSL adapter runs, so the trajectory CSV is identical (to a sub-pixel float
tolerance;
8     see the acceptance gate below) ? but WITHOUT the WSL transport:
9
10     * NO ``wslCommandMixin`` (no ``wsl -d <distro> bash -c``), NO ``conda activate`` (the
11     ``wasb`` env's python is invoked by path), NO ``/mnt/c`` path translation
12     (``to_wsl_mnt_path``). Paths are native; the video is already local on the box.
13     * ``device`` is ``cuda`` (default ? the WSL parity path), ``mps``, ``cpu``, or
``auto``
14     (M4: cuda-if-available-else-cpu ? the CPU fallback for a no-GPU box; explicit
``cuda`` stays
15     strict); threaded into ``wasb_infer.py --device`` (and so the Hydra
``runner.device`` override).
16     * weights / repo / python / scratch come from the environment first
17     (``WASB_WEIGHTS`` / ``WASB_REPO_DIR`` / ``WASB_PYTHON`` / ``WASB_SCRATCH``), then
18     ``indexing.wasb.`` config, then a default ? so a box needs no config edits.
19
20     It emits the identical ``Frame,Visibility,X,Y`` CSV (same ``{stem}__vid12_wasb.csv``
21     name as the WSL runner) and reuses the model-agnostic ``parse_trajectory_csv`` /
22     ``trajectory_to_action_windows`` from ``tracknet_runner``, so the trajectory hybrids and
23     the windowing brain are unchanged. Selected via the existing ``detector_impl`` seam
24     (``trajectory_hybrid._resolve_runner`` ? ``_build_native_runner``);
``detector_impl="native"``.
25

```

```

26     **Acceptance gate (parity, NOT strict byte-equality):** the SAME clip through the WSL
runner
27     (Windows) and this runner (Linux, ``device=cuda``) yields the same trajectory CSV + the
same
28     candidate windows **within a small float tolerance (~1e-3 px)**. cuDNN is non-
deterministic
29     ACROSS GPUs, so a sub-pixel diff is expected on different hardware ? but it is byte-
exact
30     run-to-run on a FIXED GPU. ? Validated 2026-06-20 on an RTX 2070 (via WSL): native
streaming
31     vs the cached WSL-disk CSV matched **1194/1195** frames (the lone diff a ~5e-5 px edge
32     artifact), and two native runs were byte-identical (deterministic). This is a hardware
check
33     (a real GPU + the ``wasb`` env), so the offline, subprocess-mocked suite asserts the
contract,
34     not the numbers.
35     """
36
37     import logging
38     import os
39     import shutil
40     import subprocess
41     from dataclasses import dataclass
42     from typing import List, Optional, Tuple
43
44     from backend.config.models import IndexingConfig
45     from backend.pipeline.detectors.base import DetectorRunner
46     from backend.pipeline.detectors.device import AUTO, VALID_DEVICES, DeviceContext
47
48     # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical to
WSL/TrackNet.
49     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner
50
51     logger = logging.getLogger(__name__)
52
53     # Path to the inference core we copy into the WASB repo's src/ (so it can import WASB's
54     # detectors/trackers/dataloaders + find configs/), exactly as the WSL adapter does.
55     _INFER_PY = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
56
57     # Single torch/cuda probe code, used by BOTH healthcheck and the lazy device-resolution
probe
58     # so the "cuda True/False" string they parse can never drift apart.
59     _CUDA_PROBE_CODE = (
60         "import torch; print('torch', torch.__version__, 'cuda', torch.cuda.is_available())"
61     )
62
63
64     @dataclass
65     class NativeWasbConfig:
66         """Resolved settings for a Linux-native WASB run.
67
68         Built via :meth:`from_indexing_cfg` with **environment overrides** taking precedence
69         over ``indexing.wasb.*`` config (the box sets env; no config edits needed).
``device``
70         defaults to ``cuda`` (the WSL parity path); ``stream_video`` defaults to True (the
perf win).
71         """
72
73         repo_dir: str = "~/models/WASB-SBDT" # native clone of nttcom/WASB-SBDT
74         python_bin: str = (
75             "python" # the `wasb` conda env python (invoked by path, no activate)
76         )
77         weights_path: str = "" # REQUIRED (env WASB_WEIGHTS or indexing.wasb.weights_path)
78         sport: str = "badminton"
79         device: str = "cuda" # auto | cuda | mps | cpu (auto = cuda-if-available-else-cpu)

```

```

80     scratch_dir: str = "" # frame-extraction scratch (env); "" = next to the output dir
81     timeout_sec: int = 1800 # 30 min; a hung GPU call is killed (0 = no timeout)
82     keep_frames: bool = (
83         False # retain the decoded frame cache after success (debug only)
84     )
85     # NATIVE DEFAULT = STREAMING: skip the cv2 PNG extraction that dominates a long
clip's
86     # wall-clock (measured ~50s on a 20s clip) ? bit-identical to disk (verified on a
real GPU
87     # 2026-06-20). The GPU box reads from a local NVMe so streaming is the right default
here.
88     stream_video: bool = True
89     # GPU-feed tuning (indexing.wasb.batch_size / prefetch_batches). 0 = omit the flag ?
90     # wasb_infer.py's parity defaults (batch 8, no prefetch). The cloud-serving preset
opts in.
91     batch_size: int = 0
92     prefetch_batches: int = 0
93
94     @classmethod
95     def from_indexing_cfg(cls, idx_cfg: "IndexingConfig") -> "NativeWasbConfig":
96         if isinstance(idx_cfg, dict):
97             idx_cfg = IndexingConfig(**idx_cfg)
98         w = idx_cfg.wasb
99         env = os.environ.get
100         # stream_video is tri-state in config: None/absent => native default (stream);
an
101         # explicit True/False still wins (e.g. False for a container cv2 can't seek).
102         sv = w.stream_video
103         return cls(
104             repo_dir=env("WASB_REPO_DIR") or w.repo_dir,
105             python_bin=env("WASB_PYTHON") or w.python_bin,
106             weights_path=env("WASB_WEIGHTS") or w.weights_path or "",
107             sport=w.sport,
108             device=env("WASB_DEVICE") or w.device,
109             scratch_dir=env("WASB_SCRATCH")
110             or env("RALLY_SCRATCH")
111             or env("TMPDIR")
112             or "",
113             timeout_sec=w.timeout_sec,
114             keep_frames=w.keep_frames,
115             stream_video=(True if sv is None else bool(sv)),
116             batch_size=int(w.batch_size or 0),
117             prefetch_batches=int(w.prefetch_batches or 0),
118         )
119
120
121     class NativeWasbRunner(DetectorRunner):
122         """WASB runner that executes natively on a Linux GPU box ? no WSL. See module
docstring."""
123
124         def __init__(self, cfg: NativeWasbConfig):
125             self.cfg = cfg
126             # Cached CUDA-availability probe (used to resolve device="auto"). Set by
healthcheck;
127             # lazily probed by run_predict if healthcheck was skipped. None = not yet
probed.
128             self._cuda_available: Optional[bool] = None
129
130             # --- low-level native exec (the single place tests patch)
-----
131         def _run(
132             self, argv: List[str], cwd: Optional[str] = None
133         ) -> subprocess.CompletedProcess:
134             logger.debug("native exec: %s (cwd=%s)", " ".join(argv), cwd)
135             return subprocess.run(

```

```

136         argv,
137         cwd=cwd,
138         capture_output=True,
139         text=True,
140         timeout=(self.cfg.timeout_sec or None),
141     )
142
143     # --- device resolution (M4 DeviceContext: device="auto" ? cuda-if-available-else-
cpu) -----
144     def _probe_cuda(self) -> bool:
145         """Probe ``torch.cuda.is_available()`` in the wasb env's python (a subprocess).
Any failure
146         (missing python / timeout / import error) is treated as 'no CUDA'."""
147         try:
148             res = self._run([self.cfg.python_bin, "-c", _CUDA_PROBE_CODE])
149         except (FileNotFoundError, subprocess.TimeoutExpired):
150             return False
151         return res.returncode == 0 and "cuda True" in (res.stdout or "")
152
153     def _cuda_is_available(self) -> bool:
154         """CUDA availability, cached across healthcheck + run_predict so we probe at
most once."""
155         if self._cuda_available is None:
156             self._cuda_available = self._probe_cuda()
157         return self._cuda_available
158
159     def _effective_device(self) -> str:
160         """The torch device to actually pass to ``wasb_infer.py``. Only ``auto`` needs
the CUDA
161         probe; an explicit cuda/mps/cpu resolves to itself (so ``device='cuda'`` is
unchanged)."""
162         if self.cfg.device == AUTO:
163             return DeviceContext(AUTO, self._cuda_is_available()).effective
164         return self.cfg.device
165
166     def _repo_src(self) -> str:
167         return os.path.join(os.path.expanduser(self.cfg.repo_dir), "src")
168
169     def _copy_infer_script(self) -> bool:
170         """Copy wasb_infer.py into the WASB repo src/ so it imports WASB modules + finds
configs/."""
171         repo_src = self._repo_src()
172         try:
173             os.makedirs(repo_src, exist_ok=True)
174             shutil.copy(_INFER_PY, os.path.join(repo_src, "wasb_infer.py"))
175             return True
176         except OSError as e:
177             logger.error("Failed to copy wasb_infer.py into %s: %s", repo_src, e)
178             return False
179
180     def _rm_rf(self, path: Optional[str]) -> None:
181         """Best-effort recursive delete of a native path (post-success cleanup; never
raises)."""
182         if path:
183             shutil.rmtree(path, ignore_errors=True)
184
185     def healthcheck(self) -> Tuple[bool, str]:
186         """Verify weights + repo present and the `wasb` python imports torch (and sees a
GPU on cuda)."""
187         if not self.cfg.weights_path:
188             return False, (
189                 "WASB weights path is empty ? set WASB_WEIGHTS (or
indexing.wasb.weights_path)."
190             )
191         weights = os.path.expanduser(self.cfg.weights_path)

```

```

192         if not os.path.exists(weights):
193             return False, f"WASB weights not found at {weights}"
194         repo_src = self._repo_src()
195         if not os.path.isdir(repo_src):
196             return False, (
197                 f"WASB-SBDT repo src/ not found at {repo_src} ? clone nttcom/WASB-SBDT "
198                 f"(set WASB_REPO_DIR / indexing.wasb.repo_dir).")
199         )
200         # Fail fast on a typo'd device (the device-policy seam) ? a clear error here
beats a cryptic
201         # argparse rc=2 from wasb_infer.py at run time.
202         if self.cfg.device not in VALID_DEVICES:
203             return False, (
204                 f"unknown device {self.cfg.device!r} ? valid: {'',
205                 '.join(VALID_DEVICES)})."
206             )
207         try:
208             res = self._run([self.cfg.python_bin, "-c", _CUDA_PROBE_CODE])
209         except FileNotFoundError:
210             return (
211                 False,
212                 f"python binary not found: {self.cfg.python_bin!r} (set WASB_PYTHON).",
213             )
214         except subprocess.TimeoutExpired:
215             return False, "native torch healthcheck timed out."
216         out = (res.stdout or "").strip()
217         if res.returncode != 0:
218             return (
219                 False,
220                 f"{self.cfg.python_bin}` torch import failed: {(res.stderr or
221                 out).strip()}",
222             )
223         # M4: resolve the device. Cache the probe so run_predict reuses it (no second
subprocess).
224         self._cuda_available = "cuda True" in out
225         ctx = DeviceContext(self.cfg.device, self._cuda_available)
226         if ctx.cuda_required_but_missing:
227             # Unchanged pre-M4 behaviour for an EXPLICIT cuda request: hard-fail, never
a silent
228             # slow-CPU downgrade. Set indexing.wasb.device=auto for a CPU fallback.
229             return False, f"device=cuda but torch.cuda.is_available() is False ({out})"
230         if ctx.fell_back:
231             logger.warning(
232                 "device=auto: no CUDA GPU detected (%s) ? FALLING BACK TO CPU; WASB "
233                 "inference will be slow. Set indexing.wasb.device=cuda to require a
GPU.",
234                 out,
235             )
236         return True, out
237
238     def run_predict(
239         self, video_win_path: str, output_win_dir: str, video_id: Optional[str] = None
240     ) -> Optional[str]:
241         """Run WASB natively on a video. Returns the path to the trajectory CSV, or
None.
242
243         Output artifacts are namespaced by ``video_id`` (the file MD5) exactly as the
WSL
244         runner does, and the CSV is named ``{stem}__{vid12}_wasb.csv`` ? byte-for-byte
the
245         same downstream contract, so the parity comparison is apples-to-apples.
246         """
247         output_dir = os.path.abspath(output_win_dir)
248         os.makedirs(output_dir, exist_ok=True)
249         # Normalize for a cross-platform basename (tests pass Windows paths on Linux).

```

```

248         norm = str(video_win_path).replace("\\", "/")
249         stem, _ext = os.path.splitext(os.path.basename(norm))
250         # Content-derived key: same filename + different bytes -> different key (no
cache reuse).
251         key = f"{stem}_{video_id[:12]}" if video_id else (stem or "wasb")
252         out_csv = os.path.join(output_dir, f"{key}_wasb.csv")
253
254         if not self._copy_infer_script():
255             logger.error("Could not stage wasb_infer.py into the WASB repo src/.")
256             return None
257
258         weights = os.path.expanduser(self.cfg.weights_path)
259         argv = [
260             self.cfg.python_bin,
261             "wasb_infer.py",
262             "--video",
263             os.path.abspath(str(video_win_path)),
264         ]
265         frames_dir: Optional[str] = None
266         if self.cfg.stream_video:
267             # Fast path: decode in-process, no PNG extraction. Output is bit-identical
to disk.
268             argv.append("--stream-video")
269         else:
270             scratch = self.cfg.scratch_dir or output_dir
271             frames_dir = os.path.join(scratch, f"{key}_frames")
272             argv += ["--frames_out_dir", frames_dir]
273             # M4: resolve device="auto" to the effective device (cuda-if-available-else-
cpu); an
274             # explicit cuda/mps/cpu passes through unchanged. Never sends "auto" to
wasb_infer.py.
275             device = self._effective_device()
276             argv += [
277                 "--weights",
278                 weights,
279                 "--sport",
280                 self.cfg.sport,
281                 "--device",
282                 device,
283                 "--out",
284                 out_csv,
285             ]
286             # GPU-feed tuning: pass only explicit non-zero values so the default argv (and
its
287             # byte-parity guarantee) is unchanged when the knobs are unset.
288             if self.cfg.batch_size:
289                 argv += ["--batch-size", str(self.cfg.batch_size)]
290             if self.cfg.prefetch_batches:
291                 argv += ["--prefetch-batches", str(self.cfg.prefetch_batches)]
292
293             logger.info(
294                 "Running native WASB inference (device=%s%s) on %s ...",
295                 device,
296                 " [auto?fallback]" if self.cfg.device == AUTO and device != "cuda" else "",
297                 os.path.basename(norm),
298             )
299             try:
300                 res = self._run(argv, cwd=self._repo_src())
301             except FileNotFoundError:
302                 logger.error(
303                     "python binary not found: %r (set WASB_PYTHON).", self.cfg.python_bin
304                 )
305                 return None
306             except subprocess.TimeoutExpired:
307                 logger.error(

```

```

308         "WASB native inference timed out after %ss.", self.cfg.timeout_sec
309     )
310     return None
311 if res.returncode != 0:
312     logger.error(
313         "WASB native inference failed:\n%s",
314         (res.stderr or res.stdout or "")[-2000:],
315     )
316     return None
317 if not os.path.exists(out_csv):
318     logger.error(
319         "WASB native finished but expected CSV not found at %s", out_csv
320     )
321     return None
322 logger.info("WASB trajectory CSV: %s", out_csv)
323
324 # Success ? the CSV is the durable output; the decoded frame cache is a pure
intermediate.
325 # On any earlier failure we return above WITHOUT cleaning, so the cache survives
a resume.
326 # (Streaming never materializes frames, so there is nothing to drop on that
path.)
327 if not self.cfg.keep_frames and frames_dir:
328     logger.info(
329         "Cache hygiene: removing frame cache for %s "
330         "(set indexing.wasb.keep_frames=true to retain).",
331         key,
332     )
333     self._rm_rf(frames_dir)
334     return out_csv
335
336 # Trajectory parsing + windowing are inherited from tracknet_runner (model-
agnostic).
337 parse_trajectory_csv = staticmethod(TrackNetRunner.parse_trajectory_csv)
338 trajectory_to_action_windows = staticmethod(
339     TrackNetRunner.trajectory_to_action_windows
340 )
341

```

5. assistant (2026-07-03T19:48:29.518Z)

Now let me look at the config models and presets, plus the Dockerfile for the WASB conda env.

6. user (2026-07-03T19:48:32.377Z)

```

1     """Pydantic models for the application configuration.
2
3     This module defines the canonical, validated, typed configuration for the
4     Badminton Highlight Indexer. Built-in defaults live here (single source of truth).
5
6     Unknown / typo'd config keys are rejected at load time via extra="forbid".
7     """
8
9     from __future__ import annotations
10
11     from typing import Any, List, Literal, Optional
12
13     from pydantic import BaseModel, Field, field_validator, model_validator
14
15     Sport = Literal["badminton", "tennis", "pickleball", "table_tennis", "cricket"]
16
17
18     class ValidationRelevanceConfig(BaseModel):
19         court_green_lower: list[int] = Field(default=[35, 40, 40])
20         court_green_upper: list[int] = Field(default=[85, 255, 255])

```

```

21         court_pixels_min_ratio: float = Field(default=0.05, ge=0.0, le=1.0)
22
23
24     class ValidationConfig(BaseModel):
25         min_resolution_width: int = 1280
26         min_resolution_height: int = 720
27         min_fps: float = 29.9
28         min_blur_threshold: float = 20.0
29         max_scene_cuts_per_minute: float = 0.5
30         pts_max_gap_seconds: float = Field(
31             default=10.0,
32             ge=0.0,
33             description=(
34                 "Maximum allowed keyframe gap (seconds) before flagging as spliced. "
35                 "GoPro H.264/H.265 recordings typically have GOP intervals of 1?4s; "
36                 "a gap >10s suggests edited/missing content. Mobile phones, screen "
37                 "recordings, and other encoders may use different GOP sizes ? tune "
38                 "this threshold for your recording device."
39             ),
40         )
41         relevance: ValidationRelevanceConfig = Field(
42             default_factory=ValidationRelevanceConfig
43         )
44
45
46     class TrackNetConfig(BaseModel):
47         wsl_distro: str = "Ubuntu"
48         repo_dir: str = "~/models/TrackNetV4"
49         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
50         conda_env: str = "TrackNetV4"
51         weights_path: str = ""
52         queue_length: int = 5
53         stage_in_wsl: bool = True
54         wsl_stage_dir: str = "~/clips"
55         timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
56         velocity_threshold: float = 0.02
57         # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
58         # decoded frame cache) are deleted automatically ? they are pure intermediates,
59         # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
60         # true only for debugging. On FAILURE they are always kept so a re-run can resume.
61         keep_frames: bool = False
62         # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
63         # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
64         # 0 = no extra floor (the headroom guard still applies).
65         min_free_gb: float = 0.0
66
67
68     class WasbIndexConfig(BaseModel):
69         """Typed config for the live WASB shuttle substrate (indexing.wasb).
70
71         Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
72         these knobs actually take effect ? previously this whole block was silently dropped
73         because nested config sections defaulted to extra='ignore'.
74         """
75
76         wsl_distro: str = "Ubuntu"
77         repo_dir: str = "~/models/WASB-SBDT"
78         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
79         conda_env: str = "wasb"
80         weights_path: str = (
81             "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
82         )
83         sport: str = "badminton"

```



```

84     wsl_stage_dir: str = "~/clips"
85     timeout_sec: int = 1800 # 30 min; kills a hung WSL/GPU call (0 = no timeout)
86     velocity_threshold: float = 0.02
87     # --- Linux-native runner only (detector_impl='native'); ignored by the WSL runner.
88     # Env vars override these (WASB_DEVICE / WASB_PYTHON) for a config-free GPU box.
89     # torch device for the native runner. "auto" = cuda-if-available-else-cpu (the M4
CPU fallback
90     # for a no-GPU box); "cuda" stays strict (hard-fails without a GPU ? no silent slow-
CPU downgrade).
91     device: str = "cuda" # auto | cuda | mps | cpu
92     python_bin: str = (
93         "python" # the `wasb` conda env python (invoked by path, no `conda activate`)
94     )
95     # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
96     # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates,
regenerable
97     # from the source. Set true only for debugging. On FAILURE they are kept so a re-run
resumes.
98     keep_frames: bool = False
99     # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of
the
100    # ~15x-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS,
not warns).
101    # 0 = no extra floor (the headroom guard still applies).
102    min_free_gb: float = 0.0
103    # FAST PATH ? decode the video on the fly and feed frames straight to the detector,
104    # skipping the slow per-frame PNG extraction that dominates a long clip (~46k PNGs
for a
105    # 12-min 1080p60 video, which overran the 30-min timeout). Output is bit-identical
to the
106    # PNG path (PNG is lossless) ? VERIFIED on a real GPU 2026-06-20 (native stream ==
WSL disk,
107    # 1194/1195 frames byte-identical; native run-to-run byte-identical /
deterministic).
108    # Tri-state:
109    #     None          = per-runner default ? the WSL runner keeps DISK extraction
(unchanged Windows
110    #                     behaviour); the Linux-native runner (GPU box) defaults to STREAMING
(the perf win).
111    #     True/False = force on/off for BOTH runners (e.g. set False for a container cv2
can't seek).
112    stream_video: Optional[bool] = None
113    # GPU-feed tuning for the native runner (measured on the first real L4 serving run,
114    # 2026-07-03: the default batch 8 with single-process loading left the GPU <1%
utilized ?
115    # the pipeline was CPU-feed-bound). None = wasb_infer.py's parity defaults (batch 8,
no
116    # prefetch ? byte-identical to the historical path). The cloud-serving preset opts
into
117    # the measured fast values; local/WSL behaviour is unchanged unless set explicitly.
118    batch_size: Optional[int] = None
119    # Depth of the bounded producer-thread queue that overlaps CPU-side batch assembly
120    # (decode + PIL + ToTensor) with GPU inference in streaming mode. None/0 = off.
121    prefetch_batches: Optional[int] = None
122
123
124    class FusionConfig(BaseModel):
125        """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN
P2).
126
127        Every default reproduces control behaviour: the fusion segmenter persists the
128        shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']``
is
129        true ? the person-cue features, but it does NOT gate the served handoff unless a

```

```

130     threshold is raised (`min_crossings`/`min_players` default 0 = OFF). The court
mask
131     is optional: `court_polygon=None` ? Gate B counts every detection (player-count-
only);
132     supplied ? off-court persons (spectators / adjacent courts) are excluded.
133     ""
134
135     # Which trajectory substrate seeds the windows (shuttle stays primary).
136     substrate: Literal["wasb", "tracknet"] = "wasb"
137     # Windowing velocity threshold for the fusion family (the engine reads
138     # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
139     # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
140     velocity_threshold: float = 0.02
141     # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
142     # enabled ? so the default path never imports torch / touches the GPU.
143     cue_flags: dict[str, bool] = Field(default_factory=dict)
144     # Served Gate A: drop windows whose shuttle net-crossings < this from the AI
handoff.
145     # 0 = OFF (no gating; persisted candidates are always the full superset for offline
re-scoring).
146     min_crossings: int = Field(default=0, ge=0)
147     # Served Gate B: when the person cue is on, drop windows with median in-court
players
148     # < this. 0 = OFF.
149     min_players: int = Field(default=0, ge=0)
150     # Optional court mask as normalised 0-1 [[x, y], ...] polygon. None = no mask.
151     court_polygon: Optional[list[list[float]]] = None
152     # Net line for the person two-sided-motion split (person features only ? the shuttle
153     # net-crossing gate keeps rally_gate's camera-agnostic auto-estimate).
154     net_axis: Literal["x", "y"] = "y"
155     net_position: float = Field(default=0.5, ge=0.0, le=1.0)
156
157
158     class IndexingConfig(BaseModel):
159         default_segmenter: str = "yolo_hybrid"
160         use_gpu: bool = True
161         # Detector backend for the trajectory hybrids (wasb/tracknet). "auto" = the real
162         # WSL runner (default, production); "stub" = a deterministic CPU mock (no GPU/WSL)
163         # so the whole pipeline is regression-testable on a CPU box / CI; "native" = the
164         # Linux-native WASB runner (no WSL/conda-activate; a GPU box ? M3.5). The
165         # RALLY_DETECTOR_IMPL env var overrides this. See
pipeline/detectors/{stub,native_wasb}_runner.py.
166         detector_impl: str = "auto"
167         motion_threshold: float = 0.08
168         min_segment_duration: float = 3.0
169         dead_time_merge_gap: float = 2.0
170         chunk_duration: float = 90.0
171         chunk_overlap: float = 10.0
172         detector_model: str = "fasterrcnn_resnet50_fpn"
173         detector_score_threshold: float = 0.5
174         yolo_velocity_threshold: float = 0.15
175         yolo_frame_skip: int = 3
176         yolo_tracker: str = "bytetrack.yaml"
177         yolo_prefetch_workers: int = 2
178         min_action_window_duration: float = 2.0
179         # BOUNDED WINDOWING (over-merge fix W1, RALLY_QUALITY_RESEARCH.md §6). 0 = OFF; > 0
=a window
180         # longer than this many seconds is recursively split at its largest internal
inactivity gap (?
181         # `window_min_split_gap` s ? the most likely rally boundary), so one window can't
swallow
182         # several rallies + the dead time between them. Never merges, only cuts (recall
can't drop).
183         # *** R1 MILESTONE DEFAULT-ON (cap=6): the smallest safe local-windowing flip.
Measured n=13

```

```

184      # golden, R2 tolF1: baseline?milestone (cap=6 + R4 below) ?+0.222, sign 12/0,
p=0.0005; cap=6
185      # beats cap=12 ?+0.110, sign 12/0; net over-seg guard PASS (mean merge_rate
0.651?0.161). ***
186      max_window_duration: float = 6.0
187      window_min_split_gap: float = (
188          0.5 # min internal gap (s) that qualifies as a split point
189      )
190      # HARD-CAP fallback for W1: when True, an over-long window with NO internal
inactivity gap
191      # (a continuously-active trajectory ? e.g. the real-footage mahadevpura-1 174s blob)
is
192      # chopped at its midpoint until ? max_window_duration, making the cap a true upper
bound.
193      # Only cuts (recall can't drop). OFF by default until measured/promoted.
194      window_hard_cap: bool = False
195      # R4 rally-END inactivity trim (s, 0 = OFF). After a rally the shuttle keeps moving
slowly
196      # (picked up / walked) above velocity_thresh, so the window over-extends its END
(the measured
197      # |?end| gap vs golden). Trim the post-rally tail back to the last frame whose
trailing
198      # window stays dense with FAST motion (velocity > inactivity_rally_thresh). Only
cuts.
199      # *** R1 MILESTONE DEFAULT-ON (1.5/0.20/0.30): on top of cap=6, R4 adds a small but
significant
200      # end-precision gain ? ?tolF1 +0.040, sign 9/1/3, p=0.022, |?end| 4.96?3.31s (n=13).
The W1 cap
201      # is the dominant lever; R4 is the marginal increment. See QUALITY_ITERATIONS.md
"R1". ***
202      window_inactivity_end: float = 1.5
203      inactivity_rally_thresh: float = 0.20
204      inactivity_min_density: float = 0.30
205      # R4 density DENOMINATOR fix (code-review finding B). The R4 trim's fast-fraction
denominator is
206      # the COUNT of active samples in the trailing window, which sparse tracking inflates
(one fast
207      # sample after a gap ? 100% dense ? tail held open). When True, the denominator is
the window's
208      # TEMPORAL length instead. No-op under dense tracking (sample-count == temporal-
span), so it only
209      # corrects the sparse-track case. *** P2b PROMOTED DEFAULT-ON (2026-06-29): golden-
set A/B at the
210      # SERVED operating point (n=15, 595 rallies) ? tolF1 0.353?0.364 (?tolF1 +0.010,
sign 10W/4L, NO
211      # regression), |?end| 2.97?2.72s, seg_count_ratio ?0.103 (p=0.006), split_rate
?0.010 (p=0.016),
212      # R1 over-seg guard PASS; served_regresses(eps=0.0)=False ?
promote_if_served_safe(structural)=True.
213      # See docs/RALLY_DETECTION_FIXES_PLAN.md §7/§8. ***
214      inactivity_temporal_density: bool = True
215      # R5 blob-fallback (default OFF). LAST-RESORT splitter for the continuously-active
blob: after
216      # the W1 gap-split AND the R4 inactivity trim have each had their chance, a group
STILL longer
217      # than window_blob_factor × max_window_duration (no internal gap, no rally-end
signal ? e.g.
218      # mahadevpura-1's ~65s residual ? 11× a 6s cap) is midpoint-chopped to ? the cap as
a FORCED cut
219      # (logged + flagged on the window) so the degenerate single-window outcome is
avoided and the
220      # clip is surfaced as needing rally-STATE cues, not a bigger cap. Differs from
window_hard_cap
221      # (which chops BEFORE R4 and over-segments normal clips): blob-fallback runs AFTER
R4 and only

```

```

222     # force-cuts the genuine blob (the factor gate). Only cuts (recall can't drop).
223     window_blob_fallback: bool = False
224     # Magnitude gate that keeps R5 surgical: only a group longer than this ×
max_window_duration is a
225     # blob (a normal long rally ~1-2× the cap is left alone; the mahadevpura-1 residual
is ~11×).
226     # Default 4.0 is do-no-harm on the golden corpus (rescues only the continuously-
active
227     # clips, every other clip a no-op within noise) ? see docs/QUALITY_ITERATIONS.md
"R5".
228     window_blob_factor: float = 4.0
229     log_yolo_candidates: bool = True
230     store_yolo_candidates: bool = True
231     ai_handoff_padding: float = 2.0
232     ai_max_workers: int = 5
233     # Width (px) the gemini segmenter re-encodes chunks to before upload. Stream-copied
234     # chunks of a high-bitrate (4K) source blow past the provider upload cap
235     # (backend/providers/gemini.py::MAX_UPLOAD_MB) and are refused ? observed live as
236     # "0 segments / success". Re-encoding also makes chunk boundaries frame-accurate
237     # (stream copy cuts at the nearest keyframe). 0 = legacy stream-copy at source res.
238     # Matches gemini_refine's --clip-scale-width default.
239     ai_chunk_scale_width: int = 1280
240     # FULLY-LOCAL mode (default OFF): when true, the trajectory-hybrid segmenters
241     # (wasb_hybrid / tracknet_hybrid / fusion_hybrid) skip the Phase-3 AI handoff
entirely and
242     # emit their local candidate windows AS the rallies ? no provider, no API key,
nothing
243     # uploaded. Lets the local detector run standalone (e.g. to benchmark it against the
Gemini
244     # path, or to operate with zero cloud dependency). The pure `gemini` segmenter
ignores this.
245     skip_ai_handoff: bool = False
246     # Optional debug knob: trim every video to the first N seconds before processing.
247     debug_duration: Optional[float] = None
248
249     tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
250     wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
251     fusion: FusionConfig = Field(default_factory=FusionConfig)
252
253     @field_validator("default_segmenter")
254     @classmethod
255     def segmenter_must_be_registered(cls, v: str) -> str:
256         # Lazy import: the segmenter registry imports modules that may import
257         # config, so we resolve it at validation time rather than at module load.
258         from backend.pipeline.segmenters.base import segmenter_registry
259
260         if segmenter_registry.get(v) is None:
261             available = sorted(segmenter_registry.list_available().keys())
262             raise ValueError(
263                 f"Segmenter '{v}' is not registered. Available: {'', '.join(available)}"
264             )
265         return v
266
267     @model_validator(mode="after")
268     def _chunk_step_must_be_positive(self) -> "IndexingConfig":
269         # The chunked segmenters advance by (chunk_duration - chunk_overlap); a non-
positive
270         # step makes `while t < duration: t += step` loop forever, growing memory before
any
271         # GPU work. Reject the bad config at load time instead.
272         if self.chunk_overlap >= self.chunk_duration:
273             raise ValueError(
274                 f"indexing.chunk_overlap ({self.chunk_overlap}) must be < chunk_duration
"
275                 f"({self.chunk_duration}); otherwise chunking never advances."

```

```

276         )
277         return self
278
279
280 class OutputConfig(BaseModel):
281     default_highlights_name: str = "highlights.mp4"
282     db_name: str = "sports_indexer.db"
283     # Directory where the DB, highlight reels, and processed clips are written.
284     # The CLI's --output-dir overrides this at startup (see backend/main.py:main).
285     output_dir: str = "output"
286
287
288 class WatermarkConfig(BaseModel):
289     """Branding overlay burned into each highlight clip during the per-clip re-encode
290     (backend/pipeline/stitcher/compiler.py). **On by default** ? set `enabled: false`
291     (under `stitching.watermark` in config.json) to turn it off. The text is fully
292     configurable. The concat stage stays stream-copy (-c copy)."""
293
294     enabled: bool = True
295     text: str = "Generated by Khelsutra.guru"
296     logo_path: str = "" # optional transparent PNG overlay; "" = no logo
297     font_path: str = "" # optional .ttf; "" = use the fontconfig `font` family below
298     font: str = "Sans" # fontconfig family used when font_path is empty
299     font_size_ratio: float = Field(
300         default=0.035, gt=0.0, le=0.5
301     ) # text height as a fraction of frame height
302     opacity: float = Field(default=0.85, ge=0.0, le=1.0)
303     box: bool = True # faint backing box behind the text for legibility
304     margin: int = Field(default=24, ge=0) # px inset from the chosen corner
305     position: Literal["bottom-right", "bottom-left", "top-right", "top-left"] = (
306         "bottom-right"
307     )
308
309
310 class StitchingConfig(BaseModel):
311     begin_padding: float = 0.5
312     end_padding: float = 0.5
313     use_cache: bool = False
314     min_shot_count: int = 1
315     # Long-rally reel filter (seconds; 0 = OFF). Drop detected rally pairs whose
duration
316     # (end ? start) is below this from the highlight reel, BEFORE compiling. A cleaner
317     # reel-selection knob than `min_shot_count`: shot_count is unlabeled / "Unknown" on
the
318     # fully-local no-AI path, whereas duration is derived from the rally's own
start/end. The
319     # local detector over-fires and its false positives are mostly SHORT (motion blips,
320     # background-court fragments), so this keeps the reel to the eye-catching long
rallies.
321     # OFF by default ? the detector output is unchanged, only which rallies reach the
reel.
322     min_rally_duration: float = Field(default=0.0, ge=0.0)
323     watermark: WatermarkConfig = Field(default_factory=WatermarkConfig)
324
325
326 class LoggingConfig(BaseModel):
327     """Logging knobs for the run entrypoints (see backend/utils/logging_setup.py)."""
328
329     # Third-party HTTP/RPC client loggers (httpx, httpcore, urllib3, google-genai,
330     # google.auth, grpc) emit one INFO line per request ? dozens per Gemini run, which
331     # buries our own [rally]/[compile] lines in run.log during manual QA. Default raises
332     # them to WARNING; set true to restore full chatter for request-flow debugging.
333     verbose_http: bool = False
334
335

```

```

336     class SecurityConfig(BaseModel):
337         # When set, /api/process video_path must resolve under this directory (hosted/tenant
338         # mode). Default None preserves the single-user local 'any local file' workflow.
339         allowed_video_root: Optional[str] = None
340
341         # --- Chunked upload (B2, PR-2) ---
342         # Landing zone for browser uploads (assembled from parts). Per-user subdirectories
343         # arrive with PR-3 identity scoping. ? In hosted mode, set allowed_video_root to
344         # contain upload_dir or /api/process will reject the uploaded paths.
345         upload_dir: str = "output/uploads"
346         # Whole-file cap (default 4 GB) and per-part cap. Parts stay under ~100 MB because
347         # free-tier edge proxies cap request bodies there (HOSTING_PLAN $2).
348         max_upload_mb: int = 4096
349         max_part_mb: int = 95
350         allowed_upload_extensions: List[str] = ["mp4", "mov"]
351
352         # --- M9 edge auth (Cloudflare Access, D9) ---
353         # DEFAULT-OFF: edge_auth_enabled=False ? no JWT is required/verified (the local
single-user
354         # path is unchanged). When True, /api/process requires + verifies the `Cf-Access-
Jwt-Assertion`
355         # header against the team JWKS (so a direct hit to the Cloud Run URL can't spoof the
identity
356         # header) and tags the job with the verified user_id (owner_id). The team domain
(e.g.
357         # ``https://<team>.cloudflareaccess.com``) supplies the JWKS + issuer; the AUD is
the CF Access
358         # application's audience tag. NO secrets here ? these are public identifiers.
359         edge_auth_enabled: bool = False
360         cf_team_domain: str = (
361             "" # e.g. https://<team>.cloudflareaccess.com (issuer + /certs JWKS)
362         )
363         cf_access_aud: str = "" # the CF Access application AUD tag the token must carry
364         # (CORS allow-origins is set via the RALLY_CORS_ALLOW_ORIGINS env var ? read at
import in
365         # backend/main.py so importing the app does no filesystem I/O; default [""].
Credentials stay
366         # OFF ? Cf-Access is a header, not a cookie.)
367
368
369     class StorageConfig(BaseModel):
370         """Pluggable storage backend (docs/DECOUPLED_COMPUTE/). Default ``local`` = today's
371         behaviour (filesystem paths, byte-for-byte). ``s3`` covers AWS + Cloudflare R2 + D0
Spaces
372         + MinIO (via ``endpoint_url``); ``gcs`` = Google Cloud Storage; ``gdrive`` = a
locally MOUNTED
373         Google Drive (``gdrive.mount_root``). Per-backend settings are loose dicts passed to
the constructor ?
374         **secrets are never stored here**, only endpoints / regions / file paths / source-
selectors
375         (boto3/google auth chains supply credentials). See ``backend/storage/``. """
376
377         backend: Literal["local", "s3", "gcs", "gdrive", "gdrive-api"] = "local"
378         # Output root for produced artifacts. "" => fall back to output.output_dir (non-
breaking).
379         output_root: str = ""
380         # --- Input side (COMPUTE_DECOUPLED_SERVING M2; parallel to the output
`backend`/`output_root`).
381         # Default ``local`` / "" keeps today's behaviour: a bare path / ``local://`` URI is
read IN PLACE
382         # (identity, no copy). Set these to read inputs from a bucket / mounted Drive ? a
bare/relative
383         # input name is resolved against ``input_root`` (e.g.
``input_root``="gs://bucket/videos"`` + "x.mp4"
384         # => ``gs://bucket/videos/x.mp4``) or, if ``input_root`` is empty, prefixed with

```

```

`input_backend`
385     # (`"gcs"` + "bucket/x.mp4" => `gcs://bucket/x.mp4`). An explicit scheme on the
input
386     # (`gs://?`/`s3://?`) or an absolute local path ALWAYS wins. Non-local inputs
are fetched to a
387     # scratch dir before processing (see `backend.storage.acquire_local_input`).
388     input_backend: Literal["local", "s3", "gcs", "gdrive", "gdrive-api"] = "local"
389     input_root: str = ""
390     # --- Per-video workspace (docs/PER_VIDEO_WORKSPACE.md) ---
391     # One folder per video keyed by the MD5 video_id:
<root>/[<workspace_prefix>]/<video_id>/...
392     # so a local deploy and a cloud (bucket) deploy share ONE relative layout (only the
root
393     # differs). DEFAULT-OFF: convergence is phased + flipped via a Launch Report (D5).
When OFF
394     # the legacy flat layout is used unchanged.
395     single_folder_per_video: bool = False
396     # Durable workspace root URI. "" => precedence: output_root, then output.output_dir.
397     workspace_root: str = ""
398     # Cloud namespace prefix under the root so per-video folders sit tidily beside
399     # videos/ labels/ weights/ (e.g. gs://bucket/workspaces/<video_id>). "" = root
directly.
400     workspace_prefix: str = "workspaces"
401     local: dict[str, Any] = Field(default_factory=dict)
402     s3: dict[str, Any] = Field(default_factory=dict)
403     gcs: dict[str, Any] = Field(default_factory=dict)
404     gdrive: dict[str, Any] = Field(default_factory=dict)
405     # gdrive-api settings (e.g. {"root_folder_id": "<id>"}); read via the
hyphen?underscore lookup in
406     # backend.storage.get_storage. NO secrets ? creds come from ADC over the Drive
scope.
407     gdrive_api: dict[str, Any] = Field(default_factory=dict)
408
409
410     class DeploymentConfig(BaseModel):
411         """Deployment profile ? one switch that selects a coherent bundle of compute +
storage
412         knobs (COMPUTE_DECOUPLED_SERVING M1, `docs/COMPUTE_DECOUPLED_SERVING/README.md`
§4.3).
413
414         **Pure / additive:** the empty default profile (`""`) applies NO preset, so the
pipeline
415         behaves exactly as before this section existed. A profile is opt-in (`config.json`
`deployment.profile` or the `DEPLOYMENT_PROFILE` env var); auto-detect only ever
416         *suggests* one (D15 ? cloud spend is never entered silently). Nothing in the core
reads
417         these fields yet (M1): selecting a profile works purely by pre-setting the EXISTING
knobs
418         (`indexing.detector_impl` / `skip_ai_handoff`, `storage.backend`) in the
loader.
419
420         Valid values mirror `backend/config/presets.py` (parity-pinned by a unit test)."""
421
422         # "" = no profile (today's manual knobs). Others apply presets.PROFILE_PRESETS.
423         profile: Literal["", "truly-local", "cloud-serving", "cpu-mock"] = ""
424         # Where the core runs. Set by the profile preset; overridable in config.json. Inert
in M1
425
426         # (recorded for observability; consumed by the ComputeBackend seam from M2+).
427         compute_target: Literal["", "local_gpu", "cloud_run", "cpu_mock"] = ""
428
429
430     class BillingConfig(BaseModel):
431         """Per-job cost ledger (COMPUTE_DECOUPLED_SERVING M8, D8).
432
433         **DEFAULT-OFF:** `enabled=False` ? nothing records cost (the v6 cost columns stay

```

```

NULL =
434         today's behaviour). When enabled, a job's cost is computed + stored in **USD** via
the versioned
435         ``pricebook`` DB table and **displayed in INR** at ``fx_inr_per_usd``. The default
pricebook
436         version is the seeded ``placeholder-v0`` (every rate ``0.0`` ? cost ``0``) until the
owner
437         inserts a real, calibrated version and points ``pricebook_version`` at it.
438
439         ? Even with ``enabled=True`` + a real pricebook, recorded cost stays **0** until a
follow-up
440         wires the worker to emit per-job ``usage`` (gpu/wall/egress/gemini) ? M8 lands the
ledger +
441         pricebook + the recording seam; usage emission (the metering hooks) is a separate
step.
442
443         enabled: bool = False
444         pricebook_version: str = (
445             "placeholder-v0" # the seeded $0 version; owner flips to a real one
446         )
447         fx_inr_per_usd: float = 83.0 # USD?INR display rate (Bangalore market)
448         margin: float = 0.0 # resale markup (?0); 0 = bill at cost
449
450
451     class FlywheelConfig(BaseModel):
452         """Data-flywheel harness (COMPUTE_DECOUPLED_SERVING M11, D12) ? the
consent?capture?eval?goldenize
453         loop: on a CONSENTED adult job, durably **capture** the source + rally output into
the per-video
454         workspace ? **shadow-eval** owned-vs-Gemini ? **propose promotion** to the golden
TRAINING set, so
455         the owned model climbs toward the D11 ship floor.
456
457         **DEFAULT-OFF + INERT:** ``enabled=False`` ? nothing is captured/evaluated (byte-
identical to
458         today). ? **Consent is FAKED-DENY until counsel + the consent schema land:** even
``enabled=True``
459         captures **nothing**, because ``flywheel.consent_attested`` currently returns
``False`` for every
460         job (the privacy guardrail ? D10: adults-only, explicit opt-in, derived-data + auto-
delete-row).
461         This is the PLUMBING only; the real consent gate (counsel-reviewed ToS/DPA ? consent
columns ?
462         wiring) + the owned-model + golden-label data needed to *activate* shadow-
eval/promotion are
463         owner/counsel/GPU scope. Pairs with M9-consent.
464
465         enabled: bool = False
466         workspace_prefix: str = (
467             "flywheel" # per-video workspace prefix for captured artifacts
468         )
469         shadow_eval: bool = True # run the owned-vs-Gemini shadow comparison on capture
470         min_promote_n: int = 5 # promote_if_served_safe `min_n` (the served-safe gate)
471
472
473     class AppConfig(BaseModel):
474         """Root configuration object.
475
476         All runtime configuration should be accessed via an instance of this class
477         (or the loader) rather than raw dict lookups.
478
479
480         sport: Sport = "badminton"
481         ai_provider: str = "gemini"
482         # Auto-tracking alias: 2.5-flash is legacy now that 3.5-flash is the stable

```



```

flagship;
483         # the alias never deprecates underneath us. Eval reports record the run date
alongside.
484         ai_model: str = "gemini-flash-latest"
485
486         validation: ValidationConfig = Field(default_factory=ValidationConfig)
487         indexing: IndexingConfig = Field(default_factory=IndexingConfig)
488         output: OutputConfig = Field(default_factory=OutputConfig)
489         stitching: StitchingConfig = Field(default_factory=StitchingConfig)
490         security: SecurityConfig = Field(default_factory=SecurityConfig)
491         logging: LoggingConfig = Field(default_factory=LoggingConfig)
492         storage: StorageConfig = Field(default_factory=StorageConfig)
493         deployment: DeploymentConfig = Field(default_factory=DeploymentConfig)
494         billing: BillingConfig = Field(default_factory=BillingConfig)
495         flywheel: FlywheelConfig = Field(default_factory=FlywheelConfig)
496
497         # Reject unknown keys ? typed config migration is complete (D3).
498         model_config = {
499             "extra": "forbid",
500             "validate_assignment": True,
501         }
502
503
504         # Convenience type alias for callers that want to annotate
505         Config = AppConfig
506

```

7. user (2026-07-03T19:48:33.081Z)

```

1         """Deployment-profile presets + environment auto-detection (COMPUTE_DECOUPLED_SERVING
M1).
2
3         A *deployment profile* is a single switch that applies a coherent BUNDLE of existing
4         config knobs, so the truly-local / cloud-serving / cpu-mock modes don't drift apart one
5         knob at a time (the design in ``docs/COMPUTE_DECOUPLED_SERVING/README.md`` §4.3).
6
7         Precedence (lowest ? highest), realised by the loader::
8
9             built-in model defaults < profile preset < config.json < env < worker --set
10
11         This module is **pure data + pure functions** ? no IO, and (critically) **no torch
import at
12         module load**. The loader applies a preset ONLY when a profile is EXPLICITLY selected
13         (``config.json`` ``deployment.profile`` or the ``DEPLOYMENT_PROFILE`` env var); with no
14         profile selected the loader is byte-identical to the pre-M1 behaviour (``profile=""`` ==
15         today). Auto-detect only ever **suggests** a profile ? it never auto-applies one (D15:
cloud
16         spend is never entered silently).
17         """
18
19         from __future__ import annotations
20
21         import copy
22         import os
23         from typing import Any, Mapping, Optional
24
25         # Canonical enum values, defined ONCE here. ``models.DeploymentConfig``'s ``Literal``
mirrors
26         # these and ``tests/test_config_deployment.py`` pins parity so the two can never drift.
27         PROFILES: tuple[str, ...] = ("truly-local", "cloud-serving", "cpu-mock")
28         COMPUTE_TARGETS: tuple[str, ...] = ("local_gpu", "cloud_run", "cpu_mock")
29
30         # Each profile maps 1:1 to its canonical compute_target. Used to apply the preset and to
warn
31         # when config.json explicitly overrides compute_target into a state inconsistent with

```

the profile.

```
32     COMPUTE_TARGET_FOR_PROFILE: dict[str, str] = {
33         "truly-local": "local_gpu",
34         "cloud-serving": "cloud_run",
35         "cpu-mock": "cpu_mock",
36     }
37
38     # A preset is a partial, nested override dict over the model defaults. It touches ONLY
knobs
39     # that EXIST today ? M1 is pure/additive: the new
``deployment.{profile,compute_target}`` plus
40     # ``indexing.{detector_impl,skip_ai_handoff}`` and ``storage.backend``. Later milestones
extend
41     # these bundles ? ``input_backend`` (M2) and the configurable output/workspace base (M3,
42     # Launch-Report-gated) ? deliberately NOT set here so M1 never pre-empts a gated flip.
43     PROFILE_PRESETS: dict[str, dict[str, Any]] = {
44         "truly-local": {
45             "deployment": {"profile": "truly-local", "compute_target": "local_gpu"},
46             # detector_impl stays "auto" (the runner picks WSL vs native by env on the GPU
box).
47             # Zero-cloud / privacy-first ? run fully local with no Gemini handoff.
48             "indexing": {"skip_ai_handoff": True},
49             "storage": {"backend": "local"},
50         },
51         "cloud-serving": {
52             "deployment": {"profile": "cloud-serving", "compute_target": "cloud_run"},
53             # Linux L4 (cuda); Gemini handoff ON until the owned model clears the #172 floor
(D11).
54             # wasb GPU-feed tuning: the parity defaults (batch 8, no prefetch) left the paid
L4
55             # <1% utilized on the first real serving run (2026-07-03) ? the pipeline was
56             # CPU-feed-bound. Serving opts into the measured fast values; local/WSL
unchanged.
57             "indexing": {
58                 "detector_impl": "native",
59                 "skip_ai_handoff": False,
60                 "wasb": {"batch_size": 64, "prefetch_batches": 4},
61             },
62             "storage": {"backend": "gcs"},
63         },
64         "cpu-mock": {
65             "deployment": {"profile": "cpu-mock", "compute_target": "cpu_mock"},
66             # Deterministic CPU mock ? no GPU/WSL, no provider/API. The CI / regression
profile.
67             "indexing": {"detector_impl": "stub", "skip_ai_handoff": True},
68             "storage": {"backend": "local"},
69         },
70     }
71
72
73     def is_known_profile(profile: str) -> bool:
74         """True for a non-empty, recognised profile name."""
75         return bool(profile) and profile in PROFILE_PRESETS
76
77
78     def preset_for(profile: str) -> dict[str, Any]:
79         """Return a deep COPY of the preset bundle for ``profile`` (so callers can't mutate
the
80         module-level table), or ``{}`` for ``""`` / an unknown profile."""
81         return copy.deepcopy(PROFILE_PRESETS.get(profile, {}))
82
83
84     def _truthy(val: Optional[str]) -> bool:
85         """Common env-var truthiness (``1/true/yes/on``, case/space-insensitive)."""
86         if val is None:
```

```

87         return False
88     return val.strip().lower() in {"1", "true", "yes", "on"}
89
90
91     def _cuda_available() -> bool:
92         """Best-effort CUDA probe. torch is heavy + optional, so it is imported lazily here
93         and
94         ONLY when a caller opts into ``probe_cuda``; any import/runtime failure means 'no
95         GPU'."""
96         try:
97             # Intentionally lazy + local: keeps the config-load path torch-free (torch is
98             heavy
99             # and optional). Only reached when a caller opts into probe_cuda.
100             import torch
101
102             return bool(torch.cuda.is_available())
103         except Exception:
104             return False
105
106
107     def probe_cuda_available() -> bool:
108         """Public best-effort CUDA probe (lazy torch import; ``False`` on any failure).
109         Exposed for the per-profile startup checks (``backend/config/startup.py``) so a
110         caller
111         that *wants* the GPU heads-up can pass the result in without the startup module
112         importing
113         torch itself. Best-effort by design: if torch lives only in a detector subprocess
114         env
115         (native WASB), this returns ``False`` in the main process ? which is why the startup
116         checks
117         treat a GPU mismatch as a WARNING (the runner healthcheck is the authority), never a
118         hard
119         error that could false-fail."""
120         return _cuda_available()
121
122
123     def suggest_profile(
124         env: Optional[Mapping[str, str]] = None, *, probe_cuda: bool = False
125     ) -> str:
126         """SUGGEST (never auto-apply ? D15) a deployment profile from the environment.
127
128         Signals, in priority order:
129
130         - ``K_SERVICE`` set (Cloud Run)           ? ``"cloud-serving"``
131         - ``CI`` truthy                             ? ``"cpu-mock"``
132         - ``probe_cuda`` and a CUDA GPU present ? ``"truly-local"``
133         - otherwise                               ? ``""`` (no strong suggestion)
134
135         ``probe_cuda`` defaults to ``False`` so the common config-load path stays cheap and
136         never
137         imports torch; a CLI ``--detect`` or cloud-serving startup that wants the GPU probe
138         passes
139         ``probe_cuda=True``. The result is advisory only: the caller decides whether to act
140         on it.
141
142         """
143         e = os.environ if env is None else env
144         if e.get("K_SERVICE"):
145             return "cloud-serving"
146         if _truthy(e.get("CI")):
147             return "cpu-mock"
148         if probe_cuda and _cuda_available():
149             return "truly-local"
150         return ""

```

8. user (2026-07-03T19:48:33.950Z)

```
1      # Cloud Run GPU **Job** image for the rally worker (COMPUTE_DECOUPLED_SERVING M6/M7,
ADR-012/D16).
2      #
3      # Runs `python -m backend.worker infer ?` INLINE on an L4 (detect + window + stitch co-
located so
4      # CPU/wall + egress are metered into one job ? D4). Built + pushed + verified by the
owner on the
5      # M7 real run (the $100/?10k budget, D17); CI does NOT build this (a CUDA image is too
heavy) ? the
6      # dispatch shim is proven with a fake client (tests/test_compute_backend.py).
7      #
8      # Two python stacks, deliberately separate (a framework conflict otherwise ? same split
as the
9      # native GPU box, deploy/digitalocean/gpu_setup.sh):
10     # * the MAIN app env (this repo + ffmpeg) drives the pipeline + stitching;
11     # * a conda `wasb` env (py3.8 + torch 1.11.0+cu113) runs the WASB shuttle detector as
a subprocess
12     # (the native runner shells out to $WASB_PYTHON), so the app's torch never co-
installs with it.
13     #
14     # ? WASB-C3: the pretrained badminton weights are academic-data-trained ? provenance NOT
cleared for
15     # commercial sharing. They are NOT baked into this image; stage them at runtime (a
bucket fetch or a
16     # mount) and point $WASB_WEIGHTS at them. Resolve WASB-C3 before any public, non-owner
serving.
17
18     FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04
19
20     ENV DEBIAN_FRONTEND=noninteractive \
21         PYTHONUNBUFFERED=1 \
22         PIP_NO_CACHE_DIR=1 \
23         CONDA_DIR=/opt/conda \
24         WASB_REPO_DIR=/opt/models/WASB-SBDT
25
26     # ffmpeg (stitch/decode) + DejaVu fonts (the watermark filter) + git/curl for the WASB
env build.
27     RUN apt-get update && apt-get install -y --no-install-recommends \
28         ffmpeg fonts-dejavu-core git curl ca-certificates build-essential \
29         python3 python3-pip python3-venv \
30         && rm -rf /var/lib/apt/lists/*
31
32     # --- WASB detector env (micromamba + conda-forge ? BSD-licensed, NO Anaconda ToS; P3a,
ADR-005) ---
33     # Enforces ADR-005 (every dependency ? code, data, weights, AND the build toolchain ?
stays
34     # MIT/Apache/BSD): micromamba (BSD-3) + conda-forge (community) replaces Miniconda's
Anaconda-ToS-gated
35     # `defaults` channels (the ?200-employee Terms-of-Service trap flagged in
PACKAGING_PLAN.md §3).
36     # MAMBA_ROOT_PREFIX=$CONDA_DIR (= /opt/conda) keeps the env at $CONDA_DIR/envs/wasb so
$WASB_PYTHON below
37     # is UNCHANGED. Same verified torch-1.11.0+cu113 WASB stack; the pip wheels are
unaffected by the switch.
38     ENV MAMBA_ROOT_PREFIX=/opt/conda
39     RUN curl -sSL https://micro.mamba.pm/api/micromamba/linux-64/latest \
40         | tar -xj -C /usr/local bin/micromamba \
41         && micromamba create -y -q -n wasb -c conda-forge python=3.8 \
42         && git clone --depth 1 https://github.com/nttcom/WASB-SBDT.git "$WASB_REPO_DIR" \
43         && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
44             torch==1.11.0+cu113 torchvision==0.12.0+cu113 --extra-index-url
https://download.pytorch.org/whl/cu113 \
45         && ( [ -f "$WASB_REPO_DIR/requirements.txt" ] && "$CONDA_DIR/envs/wasb/bin/python"
```

```

-m pip install -q -r "$WASB_REPO_DIR/requirements.txt" || true ) \
46     && "$CONDA_DIR/envs/wasb/bin/python" -m pip install -q \
47     hydra-core omegaconf pandas opencv-python-headless pillow scipy tqdm pyyaml
matplotlib "numpy<1.24"
48
49     # --- Main application env ---
50     WORKDIR /app
51     COPY pyproject.toml README.md ./
52     COPY requirements-trackers.txt ./
53     COPY backend ./backend
54     COPY training ./training
55     # Install with the [storage-gcs] extra: the cloud worker PULLS the gs:// input and
PUSHES the
56     # gs:// outputs/done-marker through backend.storage (google-cloud-storage is lazy-
imported there).
57     # Without it the job dies at the first gs:// fetch with `ModuleNotFoundError:
google.cloud` ? the
58     # real-Cloud-Run failure the mock-client tests can't see (the bucket I/O is faked in
tests).
59     # D13/P12: `trackers` (ByteTrack's new home) is a git-only dep (broken PyPI wheel ?
60     # roboflow/trackers#474) so it's NOT pulled by `pip install .`; install it out-of-band
before
61     # the main env, or the person-cue path crashes with ModuleNotFoundError. The immutable
git SHA is
62     # single-sourced in requirements-trackers.txt.
63     RUN python3 -m pip install --upgrade pip \
64     && python3 -m pip install -r requirements-trackers.txt \
65     && python3 -m pip install ".[storage-gcs]"
66
67     # The native runner finds the WASB env + repo by env (no `conda activate` needed);
weights come in at
68     # runtime (WASB-C3). detector_impl=native = the Linux-native WASB runner; cuda stays the
default.
69     # PYTHONUNBUFFERED: without it the worker's stdout/stderr sit in the block buffer until
process
70     # exit, so Cloud Logging shows a single end-of-run burst instead of live progress
(observed on the
71     # first real M7 runs, 2026-07-03) ? a running job looks stalled and mid-run failures log
nothing.
72     ENV PYTHONUNBUFFERED=1 \
73     WASB_PYTHON=/opt/conda/envs/wasb/bin/python \
74     WASB_REPO_DIR=/opt/models/WASB-SBDT \
75     WASB_WEIGHTS=/opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar \
76     RALLY_DETECTOR_IMPL=native
77
78     # The dispatcher (CloudRunCompute) supplies the per-execution args:
79     # infer --video-uri gs:/// --out-uri gs:/// --done-uri gs:/// --segmenter wasb_hybrid
--set ?
80     ENTRYPOINT ["python3", "-m", "backend.worker"]
81

```

9. assistant (2026-07-03T19:48:38.636Z)

I now have a complete picture. Let me verify the exact data flow point where decode feeds the detector ? specifically confirming the resize/normalize happens downstream in `ImageDataset.\_\_getitem\_\_` after `read\_image`, and check whether there's any reference to how the transform pipeline treats the PIL image (to assess whether decode-side pixel differences would even survive to the detector output). Let me search for any WASB config / transform details staged in the repo.

10. user (2026-07-03T19:48:43.520Z)

Found 19 files

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-

```

worktree\backend\pipeline\detectors\wasb_infer.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\pipeline\detectors\native_wasb_runner.py
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indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\config\models.py
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indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\tests\test_golden_fixtures.py
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indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\tests\test_golden_real_fixtures.py
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indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\eval\baselines\fixtures\README.md
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\docs\GOLDEN_REGRESSION_FIXTURES.md
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\docs\COMPUTE_DECOUPLED_SERVING\TESTING_STRATEGY.md
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\CODE_MAP.md
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\reporting\__init__.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\pipeline\segmenters\base.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\pipeline\stitcher\compiler.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\pipeline\stitcher\watermark_i18n.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\pipeline\detectors\tracknet_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\pipeline\detectors\wasb_runner.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\flywheel.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\eval\golden_fixtures.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\backend\eval\golden_real_fixtures.py
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\backend\api\uploads.py

```

11. user (2026-07-03T19:48:44.935Z)

```

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indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\deploy\cloudrun\README.md:45:- 8 vCPU / 32 GiB: decode + x264 are CPU-side (NVDEC is a
deferred D16 knob), so CPU headroom is
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\deploy\cloudrun\README.md:81:- libx264 encode (parity, D16) + NVDEC decode-only; the
image keeps ffmpeg's default libx264.
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-
worktree\docs\COMPUTE_DECOUPLED_SERVING\runlogs\README.md:24:6. **Gotchas** ? conda-ToS,

```

WASB-C3, GPU stockouts, font fallback, cold-start, NVDEC, etc.

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\COMPUTE_DECOUPLED_SERVING\README.md:43:[Omitted long matching line]

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\COMPUTE_DECOUPLED_SERVING\README.md:56:| ****ADR-012**** | GPU-native; TPU deferred; ****Cloud Run+GPU scale-to-zero = serving model****; ****NVDEC**** named as the decode lever | The serving substrate + the 7-step seed scope this expands. |

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\COMPUTE_DECOUPLED_SERVING\README.md:142:8. ? ****Codec ? D16:**** ****libx264 ENCODE**** (protect parity) + ****NVDEC DECODE-only**** (speed, no output-byte change); NVENC-encode deferred behind a config knob.

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\COMPUTE_DECOUPLED_SERVING\README.md:164:[Omitted long matching line]

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\decisions\DECISIONS.md:612:3. ****The named optimization lever is GPU-accelerated video decode (NVDEC)****, not more GPU FLOPs ? the

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\decisions\DECISIONS.md:622: helps is ****NVDEC**** (hardware decode on L4/T4) + parallel decode ? i.e. lean **into** the GPU.

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\decisions\DECISIONS.md:653:1. Containerize `backend.worker` (indexer venv + `wasb` conda env; NVDEC-capable base image).

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\decisions\DECISIONS.md:658:6. NVDEC GPU-decode to kill the extraction bottleneck.

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\decisions\DECISIONS.md:691:[Omitted long matching line]

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\decisions\DECISIONS.md:714:[Omitted long matching line]

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\past_projects\AUDIT_REPORT_khelsutra-guru_2026-07-02-COMPLETED-2026-07-02.md:577:- [P1] Khelsutra/badminton-highlight-indexer: WASB streaming decode still single-threaded (num_workers forced to 0), no NVDEC anywhere ? L4 GPU ~80% idle during detection

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\past_projects\DECOUPLED_COMPUTE\DEPLOY_REPORT_GCP_2026-06-20.md:105:- Extract long videos via the ****disk path**** (`stream\_video=false`); consider more vCPUs / ****NVDEC**** to beat

C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\scratchpad\fix-worktree\docs\archives\past_projects\DECOUPLED_COMPUTE\README.md:11:> - ****Deferred OUT** (now follow-on work):**** M3**** (serving endpoint + per-video billing) + ****M4**** (scale-to-zero + cost guard) ? the ****"Cloud Run + GPU serving" subproject**** ([ADR-012](../decisions/DECISIONS.md) seed scope). ****Full-corpus training**** (real quality, n?5) + ****WASB-C3**** weight provenance + ****NVDEC**** decode + collector ****md5**** in the export + ****DPA/consent**** for non-owner uploads all carry forward (see §06 risks + the deploy report's Next steps).

12. assistant (2026-07-03T19:48:47.756Z)

Let me read the D16 decision context and the DECISIONS lines about NVDEC to ground the parity verdict, plus confirm the exact transform flow in the WASB path.

13. user (2026-07-03T19:48:49.355Z)

```

600      **Date:** 2026-06-20 . **Status:** RATIFIED (owner, in-session). **Refines ADR-011**
(concretizes the "GCP = serving" path).
601
602      ### Decision
603
604      1. **Stay GPU-native (NVIDIA L4/T4) for all compute** ? local box, GPU VM, and the
serving path ?
605          for the foreseeable roadmap. **Do NOT invest in TPU now.** Revisit TPU **only**
if/when a large
606          *owned* transformer (Qwen3-VL Gen-2+, owner-gated) at *sustained batch* scale becomes
the dominant
607          cost, and **only after a benchmark** proves it ? never speculatively.
608      2. **Cloud Run + GPU (L4), scale-to-zero, is the supported SERVING deployment model**
for the
609          pay-per-video orchestration economics ("users pay us to orchestrate on our GCS bucket
+ VMs /
610          Cloud Run"): pay-per-request GPU-seconds = the exact per-video bill, zero idle-GPU
cost, co-located
611          with `gs://khelsutra-rally-corpus`. This is the concrete form of ADR-011's deferred
**M3/M4**.
612      3. **The named optimization lever is GPU-accelerated video decode (NVDEC)**, not more
GPU FLOPs ? the
613          measured bottleneck is video *decode*, not the model.
614
615      ### Rationale
616
617      - **Workload isn't TPU-shaped.** WASB is a tiny (5.8 MB) PyTorch CNN doing per-frame
heatmap
618          inference; the owned model is a VLM. TPUs win on large, dense, *batched* matmul at
training scale;
619          streaming per-frame CNN inference with small/dynamic batches is a poor fit.
620      - **The real bottleneck is video decode, which TPU does nothing for.** Measured on the
2026-06-20 GCP
621          L4 M2 run: cv2 PNG extraction at ~12 fps with the **GPU idle during extraction**. The
lever that
622          helps is **NVDEC** (hardware decode on L4/T4) + parallel decode ? i.e. lean *into* the
GPU.
623      - **Portability / single-runtime.** TPU needs XLA (torch_xla / JAX) ? significant
fragile NRE to port
624          WASB (torch 1.11) + the pipeline, a second runtime to maintain, and **GCP-only** ? it
would break
625          the local ? VM ? Cloud Run parity the DECOUPLED storage/compute seams exist to
provide. One portable
626          PyTorch+CUDA image that runs identically everywhere is worth more than peak FLOPs
here.
627      - **Economics.** Cloud Run+GPU scale-to-zero meters per request = clean per-video
billing; TPU pricing
628          (v5e pods / queued / committed-use) targets sustained training and its regional
availability is
629          *narrower* than GPU (the M2 run already hit a Mumbai GPU stockout ? landed in
Singapore; TPU would be
630          worse).
631
632      ### Consequences
633
634      - The **deferred M3 (inference serving endpoint + per-video billing) and M4 (scale-to-
zero + cost
635          guard)** from ADR-011 become a **new tracked subproject ? "Cloud Run + GPU serving"
? to be spun
636          up (as a `docs/PROJECT_DOC_TEMPLATE.md`-based doc) **AFTER `docs/DECOUPLED_COMPUTE/`
is archived**
637          (owner sequencing). Seed scope below. **? REALIZED in ADR-014 (2026-06-21):** the
subproject is
638          `docs/COMPUTE_DECOUPLED_SERVING/`, **renamed** to reflect that **truly-local is a co-
equal profile**

```



```

639     (not just cloud serving); ADR-014 expands this 7-point seed into the M0?M10 plan.
640 - **First subproject work item = containerize `python -m backend.worker`** for Cloud
Run. This is the
641     known hard part: WASB's torch-1.11 conda env makes the image non-trivial (two runtimes
? the
642     indexer venv + the `wasb` conda env). The decoupled storage/compute seams + the native
runner
643     (`detector_impl="native"`, PR #256) make the *code* portable; packaging is the gap.
644 - Keep the **storage/compute decoupling seams** intact so the SAME container runs local
/ GPU VM /
645     Cloud Run (the per-video workspace convention, in progress, reinforces this).
646 - **Edge auth is a hard prerequisite** before any public serving (engine API is CORS
`*`, no auth) ?
647     carried from the appliance track.
648 - **Region/capacity reality:** GPU stockouts are common (Mumbai was fully stocked-out on
2026-06-20);
649     scale-to-zero + multi-region/flexible-zone selection is part of why Cloud Run+GPU
fits.
650 - TPU is explicitly **out of scope**; this ADR records *why* so it isn't relitigated.
651
652 ##### Seed scope for the "Cloud Run + GPU serving" subproject (formalize post-archival)
653 1. Containerize `backend.worker` (indexer venv + `wasb` conda env; NVDEC-capable base
image).
654 2. Cloud Run + GPU (L4) service: `infer` over a `gcs://` workspace; scale-to-zero.
655 3. Per-video cost ledger (on the `jobs` table) = the per-customer bill (M3).
656 4. Cost guard + concurrency caps (M4).
657 5. Edge auth (tenancy boundary) before public exposure.
658 6. NVDEC GPU-decode to kill the extraction bottleneck.
659 7. DPA / consent gate for non-owner uploads (productization, not compute).
660
661 ## ADR-013: Drop DigitalOcean for compute ? GCP is the SOLE compute stack (training +
serving)
662
663 **Date:** 2026-06-20 . **Status:** RATIFIED (owner, in-session: "drop DO and document
it? record it and bake it for all future work"). **Supersedes the DO=training split in ADR-011;
refines ADR-010 + ADR-012.**
664
665 ### Decision
666
667 1. **DigitalOcean is DROPPED as a compute provider.** **GCP is the sole compute stack**
for BOTH training and serving. (ADR-011's "DO=training, GCP=serving" split is retired ? GCP does
both. The DECOUPLED scope decision in ADR-011 ? M0?M2 + M3.5 ? still stands.)
668 2. **Standing default for ALL future compute work = GCP** (a GPU VM today via
`deploy/gcp/`; Cloud Run+GPU for serving per ADR-012). Do **not** re-evaluate DO/Paperspace for
compute without a new ADR.
669 3. The `$200` DO credits, if used at all, go **only to non-compute** that DO bills
cleanly (CPU droplets / a site host / Spaces storage / bandwidth) ? **never GPU**.
670
671 ### Rationale (researched 2026-06-20, against DO's own docs)
672
673 - **DO GPU is neither enabled on the account nor a clean free-credits path.** The DO API
`/v2/sizes` returns **0 GPU sizes** (GPU Droplets not enabled); GPU Droplets are **North-
America-only** (NYC2/TOR1/ATL1/RIC1/AMS3 ? no India/Singapore); and DO's own docs state that for
**newer customers DO charges the team's primary payment method** when GPU Droplets are created
(anti-abuse) ? so GPU spend hits the **card**, not the promo credit. The ADR-011 premise ("200
credits fund DO training") does **not** hold.
674 - **Paperspace (now DO Gradient) free GPU is subscription-gated** (a paid "G"/Growth
plan; high-end A100/H100 needs the $39/mo Growth plan), and Paperspace bills **separately** from
DO Droplet credits ? the $200 credit doesn't fund it either.
675 - **GCP works today:** quota granted (`GPUS_ALL_REGIONS=1`), bucket + SA + `$100` budget
provisioned, and M1/M2/M3.5 ran on an L4 for ~$12/run with no subscription. Real money, but
trivial per run ? and **one** portable PyTorch+CUDA stack (local ? GPU VM ? Cloud Run), no
second provider to maintain.
676
677 ### Consequences

```

678

```
679 - **Compute provisioning = `deploy/gcp/` only.** `SETUP_NEW_MACHINE.md` repointed to
GCP; `deploy/digitalocean/README.md` marked **DEPRECATED** ; `HOSTING_PLAN.md`'s "All-
DigitalOcean" compute option retired.
680 - **`deploy/digitalocean/` scripts are KEPT** ? `bootstrap.sh` / `mount_scratch.sh` /
`gpu_setup.sh` are **provider-agnostic** Linux-GPU setup the GCP runbook reuses; only the DO-
specific *provisioning* (doctl droplet create, Spaces) is dropped. (Optional later cleanup: move
them to `deploy/common/`.)
681 - The shared DO CPU droplet **`claudes-do-rally-test`** (another session's testbed) is
**owner-coordinated** cleanup, separate from this decision; the freshly-rotated **DO API token**
can be retired once that droplet is gone.
682 - **Storage stays bucket-abstracted** (`gcs://` primary). The `s3` backend still speaks
any S3-compatible store, so **DO Spaces remains a *possible* storage target** even though DO
*compute* is dropped.
683 - **Capacity reality:** GCP GPU stockouts are common (asia-south1/Mumbai was fully
stocked-out 2026-06-20 ? the M2 run landed an L4 in asia-southeast1/Singapore; weights still
land in the asia-south1 bucket). Sweep zones/regions; scale-to-zero (ADR-012) is part of why
Cloud Run+GPU fits.
684 - **DECOUPLED_COMPUTE** archives on M2 completion with a note that the DO leg was
**dropped** (this ADR), not "delivered on DO".
685
686 ## ADR-014: Compute-decoupled rally serving ? one pure core, profile-selected compute +
storage
687
688 **Date:** 2026-06-21 . **Status:** ? **RATIFIED** (owner, 2026-06-21 ? M0 design sign-off;
build M1+). **Refines ADR-011 + ADR-012** (it *realizes* their deferred M3/M4; it does **not**
supersede them); **builds on ADR-013** (GCP sole compute stack). Full design:
`docs/COMPUTE_DECOUPLED_SERVING/`.
689
690 ### Evolution / lineage (trace the idea)
691 `ADR-005` (permissive licensing ? engine + image stay MIT/Apache/BSD) . `ADR-008`
(train==serve featurizer single-sourcing ? the parity discipline this extends to local?cloud) .
`ADR-009` (**local-first delivery** ? origin of the truly-local profile) . `ADR-010` (DECOUPLED
DO?GCP) . `ADR-011` (**deferred M3 serving + M4 billing** ; overrode "rent nothing / not a
service"; $06 gates = DPA/consent, collector `md5`, WASB-C3) . `ADR-012` (**Cloud Run+GPU =
serving model** ; NVDEC lever) . `ADR-013` (**GCP sole compute stack**) . **ADR-014** (this).
Non-ADR: realizes `PLATFORM_ARCHITECTURE.md`'s `ComputeBackend`/`StorageBackend` registries; the
truly-local profile = `VIDEO_LOCALITY_MODEL.md` L0 + `DESKTOP_APP_PLAN.md` `LocalDesktop` ;
informed by the archived `DEPLOY_REPORT_GCP_2026-06-20` (cross-GPU sub-pixel parity finding).
692
693 ### Decision
694
695 1. The rally engine is a **PURE** function of (input bytes + config) ? outputs**, in three
deterministic stages ? **DETECT** (`DetectorRunner.run_predict`?CSV), **WINDOW**
(`trajectory_to_action_windows`, CPU-pure), **STITCH** (`VideoCompiler.compile_highlights`,
ffmpeg) ? that are **NEVER** branched on the deployment profile**.
696 2. A **`ComputeBackend` / deployment profile** decides **WHERE** the core runs (local
process vs a Cloud Run GPU job); a **`StorageBackend`** decides **WHERE** I/O comes from** (local
FS / USB / mounted Google Drive / bucket). **Both are selected by ONE startup/setup config**
(`deployment.profile` + `compute_target`), realizing the PLATFORM_ARCHITECTURE
`local|hosted|byo_worker` registry.
697 3. Ship at least two profiles: **truly-local** (local GPU + **in-process stitch**, no
cloud) and **cloud-serving** (a Cloud Run GPU job runs detect **AND** stitch), plus **cpu-mock**
for CI and a reserved, **deferred** `byo_worker` hybrid.
698 4. In cloud-serving, **STITCH** runs on Cloud Run** (co-located with detect) so its
CPU/wall-time + egress are **metered** into a per-job cost ledger alongside `gpu_active_seconds`
? stitching is compute-intensive and its cost must be accounted for. (Truly-local stitch is
intentionally **un**metered.)
699 5. **Parity** is enforced at **WINDOW + STITCH** (byte-exact for a fixed trajectory);
**DETECT** is byte-exact only **same-GPU** ? cross-GPU is sub-pixel-different (deploy report
2026-06-20), so it is judged at the **rally/window level**, never CSV bytes.
700 6. Land **smallest-safe-first, one PR per milestone (M0?M10)** , **DEFAULT-OFF** where a
change touches core callers (e.g. the configurable output base), each guarded by golden-fixtures
+ a **profile-parity test** ; default flips ship a **Launch Report** .
701
```

```

702     ### Rationale
703
704     - The seams already exist: `_resolve_runner` is the single compute seam, the
`StorageBackend` ABC + 4 backends + the worker fetch/put are built, and the stub detector
(`replay_csv`) gives a **$0 deterministic** core. Realizing a profile system is **composition,
not a rewrite**.
705     - Separating **WHERE** (ComputeBackend, coarse, per-job) from **HOW** (DeviceContext,
fine, per-GPU) from **WHAT-bytes** (StorageBackend) keeps each axis independently testable and
avoids the main trap ? conflating them.
706     - **Metering stitch on Cloud Run** is the only way the owner's real serving cost (GPU +
CPU re-encode + egress) is attributable per job; stitch-on-laptop hides it.
707     - Aligns with ADR-012 (GPU-native) + ADR-013 (GCP sole stack) + the 2026-06-20 deploy
report; the truly-local profile preserves the ADR-009 privacy/offline self-host story.
708
709     ### Consequences
710
711     - **Positive:** one core, provably identical across modes; CI proves the decoupling for
**$0** (stub + storage fakes + a profile-parity test); new compute targets (byo_worker,
SkyPilot, Modal) slot into the enum without core changes; **cost becomes visible per job**
(incl. stitch).
712     - **Cost/new surface:** a new `deployment`+`compute_target` config surface, a Cloud Run
**dispatch shim**, a **containerized GPU+ffmpeg worker image**, a **cost-ledger migration**, and
**edge auth** must all be built + maintained. Cross-GPU detection is **not** byte-identical ?
parity asserted at the window level. The `byo_worker` tier carries a real ops/support burden
(deferred, power-user only).
713     - **Risk-managed:** every core-touching change is default-OFF + golden/parity-gated +
Launch-Reported; real-GPU runs are **owner-side** with a budget guard and a posterity
DEPLOY_REPORT (`docs/COMPUTE_DECOUPLED_SERVING/runlogs/`).
714     - **Owner gates ? the 5 gating ones LOCKED 2026-06-21** (project doc $2 **D7?D12**): D7
cold-start = **scale-to-zero** ; D8 pricebook = **versioned DB table, USD-internal/INR-display** ;
D9 edge-auth = **Cloudflare Access** (verify Cf-Access JWT, lock ingress); D10 = **free-tier
data-for-reels trade** (adults-only, derived-data, counsel-gated to M9); D11 quality-floor =
**Gemini-ON until the owned model clears the floor** + D12 the **data-flywheel harness**
(capture?store?eval?goldenize). Plus **3 more locked at M0 sign-off (2026-06-21, $2 D13?D15):**
**D13** profile is user-switchable anytime (advertised feature; equivalent output, not bit-
identical cross-GPU); **D14 NORTHSTAR** = mobile/cloud-native point-at-Google-Drive ? reel back
in Drive next to the source (implies a new **`gdrive-api` OAuth** backend = M12; resolves the
gdrive-in-cloud question); **D15** auto-detect SUGGESTS, the user CONFIRMS ? execution (esp.
cloud spend) is never a surprise. Still open (non-gating): stitch co-located vs split; NVDEC vs
libx264; the `<2GB` cap; the dir name.
715

```

14. user (2026-07-03T19:48:50.742Z)

```

1     """WASB shuttle-trajectory runner (WSL2 subprocess adapter).
2
3     Windows-side counterpart to backend/detectors/wasb_infer.py. It:
4         1. stages the video into the WSL filesystem (fast local reads, not /mnt),
5         2. copies wasb_infer.py into the WASB repo's src/ dir,
6         3. runs it in the WSL `wasb` conda env -> a Frame,Visibility,X,Y trajectory CSV,
7         4. reuses the (model-agnostic) trajectory->action-window logic from tracknet_runner.
8
9     So the only WASB-specific bits live here; the windowing + CSV parsing are shared
10    with the TrackNet path. Verified inference core: wasb_infer.py (see its docstring).
11    """
12
13    import logging
14    import os
15    import shlex
16    import subprocess
17    from dataclasses import dataclass
18    from typing import Optional, Tuple
19
20    from backend.config.models import IndexingConfig
21    from backend.pipeline.detectors.base import (

```

```

22         DetectorRunner,
23         WslCommandMixin,
24         wsl_tilde_quote,
25     )
26
27     # Reuse the model-agnostic helpers ? trajectory shape + windowing are identical.
28     # parse_trajectory_csv / trajectory_to_action_windows are @staticmethods on
TrackNetRunner.
29     from backend.pipeline.detectors.tracknet_runner import TrackNetRunner, to_wsl_mnt_path
30
31     logger = logging.getLogger(__name__)
32
33     # Windows path to the inference wrapper we stage into WSL.
34     _INFER_WIN = os.path.join(os.path.dirname(os.path.abspath(__file__)), "wasb_infer.py")
35
36
37     @dataclass
38     class WasbConfig:
39         """Resolved settings for a WASB WSL run (built from config.json ->
indexing.wasb)."""
40
41         distro: str = "Ubuntu"
42         repo_dir: str = "~/models/WASB-SBDT"
43         conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
44         conda_env: str = "wasb"
45         weights_path: str = (
46             "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
47         )
48         sport: str = "badminton"
49         wsl_stage_dir: str = "~/clips"
50         timeout_sec: int = 1800 # 30 min; a hung GPU/conda call is killed (0 = no timeout)
51         keep_frames: bool = (
52             False # retain staged video + frame cache after success (debug only)
53         )
54         min_free_gb: float = 0.0 # absolute floor (GB) atop the ~15x-source BLOCK guard (0
= off)
55         # FAST PATH (default OFF until verified): decode the staged video on the fly inside
56         # WSL and feed frames straight to the detector, skipping the slow per-frame PNG
57         # extraction that dominates a long clip. Output is bit-identical to the PNG path
58         # (PNG is lossless), but this stays opt-in until the owner's side-by-side confirms
it.
59         stream_video: bool = False
60

```

15. assistant (2026-07-03T19:49:00.245Z)

I now have everything I need. Let me capture the one remaining critical detail: the exact per-pixel path from decode to detector, to nail the parity verdict. The key facts from the code:

- The decode ? detector data flow** (wasb_infer.py):
 - CPU decode: ``cv2.VideoCapture.read()`` ? BGR uint8 ndarray (``make_video_frame_loader``, lines 743-783)
 - ``_bgr_to_pil_rgb`` (lines 328-340): ``cv2.cvtColor(frame_bgr, COLOR_BGR2RGB)`` ? ``Image.fromarray`` ? PIL RGB
 - Fed into WASB's ``ImageDataset.__getitem__`` via the ``read_image`` shim (lines 361-397), which then applies ``build_img_transforms(cfg)`` ? resize to ``input_wh`` + ToTensor + normalize (lines 504-505, 581)
 - Then ``detector.run_tensor(imgs, trans)`` (line 646)
- The parity guarantee** rests on: ``frame_bgr ? cv2.imwrite(PNG) ? Image.open(PNG).convert('RGB')`` being **byte-identical** to ``cv2.cvtColor(frame_bgr, BGR2RGB)`` ? verified ``max|diff|=0`` because PNG is lossless. The pixels the detector sees are **the exact bytes cv2 decoded**.

I have enough to give a definitive, code-grounded verdict.

16. user (2026-07-03T19:49:41.438Z)

Structured output provided successfully